Near-field singularity extraction for rooftop basis functions on uniform rectangular grids

Contents

1 Introduction 2

2 Setting: cells, rooftops and moments 2

3 Singularity extraction on the touching cells 6

4 Reduction to two dimensions 7

5 Mixed antiderivatives 11

6 Closed forms for the touching cells 14

A The mixed antiderivatives $G_{mn}^{(p)}$ 18

B The touching moments 20

References 37

1 Introduction

This note continues the bachelor thesis [1]. There the Galerkin matrix of the electric field integral equation (EFIE) with rooftop basis functions on a uniform rectangular grid is assembled from three Toeplitz generators. Their entries are four-dimensional integrals of the Green's function

$$g(R) = \frac{\mathrm{e}^{-\mathrm{i}kR}}{R}, \qquad R = \|\boldsymbol{r} - \boldsymbol{r}'\|, \tag{1}$$

over pairs of cells, weighted by the linear profiles of the rooftops (the factor $1/(4\pi)$ is kept outside). They are integrated with a tensor Gauss–Legendre (GL) rule of order q, and the singularity is extracted on the coincident cell only (Entry 2.1 in Section 2.6 and the appendix of [1]).

Cells that share an edge or only a vertex meet $R = 0$ as well, on the boundary of the integration domain. There the GL error does not decrease with the order: the generator entries keep a relative error of about 10^{-3} for cells of $\lambda/10$ at every order between 3 and 16, and extraction on the coincident cell alone changes nothing measurable. The note therefore extracts the non-smooth terms of (1),

$$\frac{1}{R}, \qquad -\frac{k^2}{2}R, \qquad \frac{k^4}{24}R^3, \tag{2}$$

on all nine touching cells and gives the closed forms of every static integral this needs.

Three terms are enough. With the three terms the entries reach 10^{-13}–10^{-14} at the order 12–16, and the computed current is at the rounding floor of double precision. A fourth term (R^5) would act below that floor, so no further extraction is needed.

The way to the closed forms. Section 2 expresses the three generators through seven cell-pair moments, so that only integrals over pairs of cells with linear weights are left. Section 3 splits the Green's function into the three extracted terms and a smooth remainder; the remainder stays with the GL rule, and the extracted terms leave the static moments with the kernels R^{-1}, R and R^3. Section 4 reduces these four-dimensional integrals to two dimensions, and Section 5 gives the antiderivatives of the resulting integrands. Section 6 puts them together: every static moment of a touching cell pair is a short sum of antiderivative values. The note is written as a manual to the implementation; all closed forms are listed in Appendices A and B in the form of the thesis appendix.

Conventions. The time convention is $\mathrm{e}^{\mathrm{i}\omega t}$. The notation follows the thesis [1].

2 Setting: cells, rooftops and moments

Definition 2.1 (Uniform grid). Let $a, b > 0$ be the cell sizes in the x and y directions. For $(m, n) \in \mathbb{Z}^2$ the cell is

$$\Omega_{mn} = [ma, (m+1)a] \times [nb, (n+1)b] \subset \mathbb{R}^2. \tag{3}$$

A point $\boldsymbol{r} = (x, y) \in \Omega_{mn}$ has the local coordinates

$$\xi(\boldsymbol{r}) = \frac{x}{a} - m \in [0, 1], \qquad \eta(\boldsymbol{r}) = \frac{y}{b} - n \in [0, 1]. \tag{4}$$

For an offset $\boldsymbol{d} = (d_x, d_y) \in \mathbb{Z}^2$ we write $\Omega_{\boldsymbol{d}} = \Omega_{d_x d_y}$. The two cells $\Omega_{\mathbf{0}}$ and $\Omega_{\boldsymbol{d}}$ share at least one point if and only if $\boldsymbol{d} \in \mathcal{T} = \{-1, 0, 1\}^2$. We call these offsets touching (Figure 1): $\boldsymbol{d} = \mathbf{0}$ is the coincident cell, the four offsets $(\pm 1, 0)$, $(0, \pm 1)$ share an edge and the four offsets $(\pm 1, \pm 1)$ a vertex.

Definition 2.2 (Rooftop functions). The x-directed rooftop $\boldsymbol{\Psi}^x_{mn}$ is supported on $\Omega_{mn} \cup \Omega_{m+1,n}$ and

$$\boldsymbol{\Psi}^x_{mn}(\boldsymbol{r}) = \hat{\boldsymbol{x}} \cdot \begin{cases} \xi(\boldsymbol{r}), & \boldsymbol{r} \in \Omega_{mn} \quad \text{(rising wedge)}, \\ 1 - \xi(\boldsymbol{r}), & \boldsymbol{r} \in \Omega_{m+1,n} \quad \text{(falling wedge)}. \end{cases} \tag{5}$$

Its normal component on the common edge is one, and its divergence is $+1/a$ on Ω_{mn} and $-1/a$ on $\Omega_{m+1,n}$. The y-directed rooftop $\boldsymbol{\Psi}^y_{mn}$ is defined in the same way with η, b and $\Omega_{m,n+1}$.

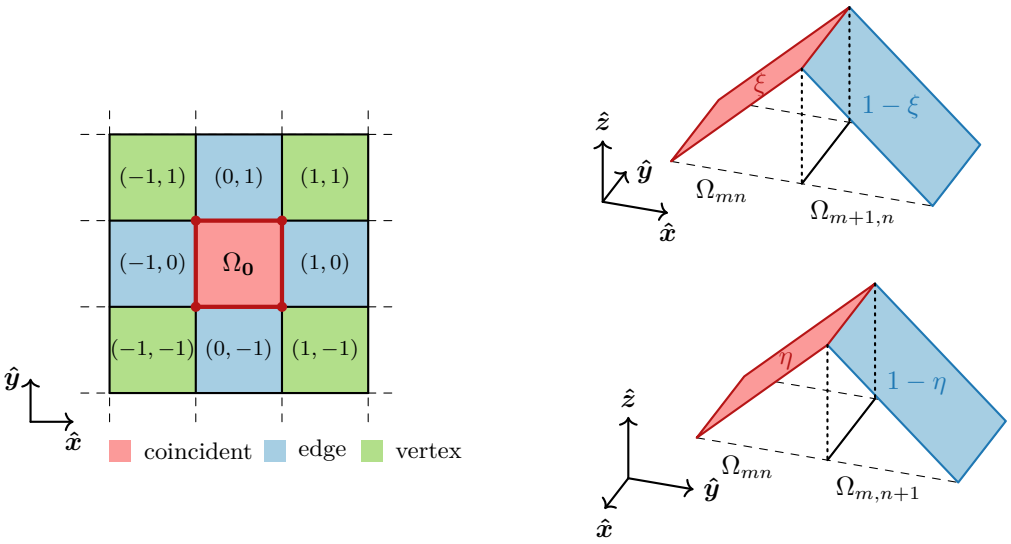


Figure 1: Left: the reference cell $\Omega_{\mathbf{0}}$ and its eight touching neighbours $\Omega_{\boldsymbol{d}}$, labelled by the offset $\boldsymbol{d} \in \mathcal{T}$. The points with $R = 0$ that the two cells have in common are drawn in red: all of $\Omega_{\mathbf{0}}$ for the coincident cell, an edge for an edge neighbour and a vertex for a vertex neighbour. Right: the x-directed rooftop $\boldsymbol{\Psi}^x_{mn}$ (top), split into its rising wedge ξ on Ω_{mn} and its falling wedge $1-\xi$ on $\Omega_{m+1,n}$, and the y-directed rooftop $\boldsymbol{\Psi}^y_{mn}$ (bottom) with the wedges η on Ω_{mn} and $1 - \eta$ on $\Omega_{m,n+1}$.

Entry 2.1 (Galerkin entries). With (1), the Galerkin entry of the EFIE between rooftops $\boldsymbol{\Psi}_p$ and $\boldsymbol{\Psi}_q$ is [1]

$$\begin{aligned} Z_{pq} = \frac{\mathrm{i} Z_0}{4\pi k} \Big(& k^2 \iint \boldsymbol{\Psi}_p(\boldsymbol{r}) \cdot \boldsymbol{\Psi}_q(\boldsymbol{r}') \, g(R) \, \mathrm{d}A' \, \mathrm{d}A \\ & - \iint \nabla \cdot \boldsymbol{\Psi}_p(\boldsymbol{r}) \, \nabla' \cdot \boldsymbol{\Psi}_q(\boldsymbol{r}') \, g(R) \, \mathrm{d}A' \, \mathrm{d}A \Big). \end{aligned} \tag{6}$$

By translation invariance the entries depend only on the offset of the two rooftops. They are generated by

$$A^{xx}(\boldsymbol{d}) = \iint \boldsymbol{\Psi}^x_{\mathbf{0}} \cdot \boldsymbol{\Psi}^x_{\boldsymbol{d}} \, g, \qquad A^{yy}(\boldsymbol{d}) = \iint \boldsymbol{\Psi}^y_{\mathbf{0}} \cdot \boldsymbol{\Psi}^y_{\boldsymbol{d}} \, g, \qquad \Phi(\boldsymbol{d}) = \int_{\Omega_{\mathbf{0}}} \int_{\Omega_{\boldsymbol{d}}} g. \tag{7}$$

The divergences are piecewise constant, so the scalar-potential parts of all blocks are finite differences of Φ; for example $\iint \nabla\cdot\boldsymbol{\Psi}^x_{\mathbf{0}}\,\nabla'\cdot\boldsymbol{\Psi}^x_{\boldsymbol{d}}\,g = a^{-2}\big(2\Phi(\boldsymbol{d})-\Phi(\boldsymbol{d}+\boldsymbol{e}_x)-\Phi(\boldsymbol{d}-\boldsymbol{e}_x)\big)$.

Definition 2.3 (Cell-pair moments). Let $K:(0,\infty)\to\mathbb{C}$ satisfy $K(\|\cdot\|)\in L^1_{\mathrm{loc}}(\mathbb{R}^2)$. For $\boldsymbol{d}\in\mathbb{Z}^2$ and $\alpha,\beta\in\{0,1\}$ define

$$\begin{aligned} M^{[K]}_{\alpha\beta}(\boldsymbol{d}) &= \frac{1}{(ab)^2}\int_{\Omega_{\mathbf{0}}}\int_{\Omega_{\boldsymbol{d}}} \xi(\boldsymbol{r})^\alpha\,\xi(\boldsymbol{r}')^\beta\,K\big(\|\boldsymbol{r}-\boldsymbol{r}'\|\big)\,\mathrm{d}A(\boldsymbol{r}')\,\mathrm{d}A(\boldsymbol{r}),\\ N^{[K]}_{\alpha\beta}(\boldsymbol{d}) &= \frac{1}{(ab)^2}\int_{\Omega_{\mathbf{0}}}\int_{\Omega_{\boldsymbol{d}}} \eta(\boldsymbol{r})^\alpha\,\eta(\boldsymbol{r}')^\beta\,K\big(\|\boldsymbol{r}-\boldsymbol{r}'\|\big)\,\mathrm{d}A(\boldsymbol{r}')\,\mathrm{d}A(\boldsymbol{r}), \end{aligned} \tag{8}$$

where $\xi(\boldsymbol{r}')$ and $\eta(\boldsymbol{r}')$ are the local coordinates in $\Omega_{\boldsymbol{d}}$. For $K(R)=R^p$ we write $M^{(p)}_{\alpha\beta}(\boldsymbol{d})$ and $N^{(p)}_{\alpha\beta}(\boldsymbol{d})$. Since $N_{00}=M_{00}$, there are seven moments, M_{00}, M_{10}, M_{01}, M_{11}, N_{10}, N_{01} and N_{11}.

The moments carry all the geometry. The next lemma lists their symmetries.

Lemma 2.1 (Symmetries of the moments). Let K be admissible, $\boldsymbol{d}=(d_x,d_y)\in\mathbb{Z}^2$ and $\alpha,\beta\in\{0,1\}$. Then:

(i) $M^{[K]}_{\alpha\beta}(d_x,-d_y)=M^{[K]}_{\alpha\beta}(d_x,d_y)$;

(ii) reflection in x:

$$\begin{aligned} M^{[K]}_{00}(-d_x,d_y) &= M^{[K]}_{00}(\boldsymbol{d}),\\ M^{[K]}_{10}(-d_x,d_y) &= \big(M^{[K]}_{00}-M^{[K]}_{10}\big)(\boldsymbol{d}),\\ M^{[K]}_{01}(-d_x,d_y) &= \big(M^{[K]}_{00}-M^{[K]}_{01}\big)(\boldsymbol{d}),\\ M^{[K]}_{11}(-d_x,d_y) &= \big(M^{[K]}_{00}-M^{[K]}_{10}-M^{[K]}_{01}+M^{[K]}_{11}\big)(\boldsymbol{d}); \end{aligned} \tag{9}$$

(iii) $N^{[K]}_{\alpha\beta}(d_x,d_y;a,b)=M^{[K]}_{\alpha\beta}(d_y,d_x;b,a)$ (the cell sizes written explicitly);

(iv) $M^{[K]}_{\alpha\beta}(\boldsymbol{d})=M^{[K]}_{\beta\alpha}(-\boldsymbol{d})$.

Proof. The kernel depends only on $\|\boldsymbol{r}-\boldsymbol{r}'\|$, which is invariant under reflections, translations and the interchange of $\boldsymbol{r}$ and $\boldsymbol{r}'$.

(i) The reflection $y\mapsto b-y$ maps $\Omega_{\mathbf{0}}$ onto itself and $\Omega_{(d_x,-d_y)}$ onto $\Omega_{(d_x,d_y)}$. It leaves both local x-coordinates unchanged.

(ii) The reflection $x\mapsto a-x$ maps $\Omega_{\mathbf{0}}$ onto itself with $\xi\mapsto 1-\xi$, and $\Omega_{(-d_x,d_y)}$ onto $\Omega_{(d_x,d_y)}$ with $\xi'\mapsto 1-\xi'$. Indeed, $[-d_xa,(1-d_x)a]$ goes to $[d_xa,(d_x+1)a]$ and $x'/a+d_x\mapsto 1-(x'/a+d_x)$. Expand $(1-\xi)^\alpha(1-\xi')^\beta$.

(iii) Exchange the roles of x and y.

(iv) Interchange $\boldsymbol{r}\leftrightarrow\boldsymbol{r}'$ and translate by $-\boldsymbol{d}$; the cell $\Omega_{\boldsymbol{d}}$ becomes $\Omega_{\mathbf{0}}$ and $\Omega_{\mathbf{0}}$ becomes $\Omega_{-\boldsymbol{d}}$.

□

Proposition 2.1 (Generators from moments). For $\boldsymbol{d}=(d_x,d_y)$ with $d_x\ge 0$,

$$\begin{aligned} \frac{A^{xx}(\boldsymbol{d})}{(ab)^2} &= M_{11}(\boldsymbol{d})+\big(M_{00}-M_{10}-M_{01}+M_{11}\big)(\boldsymbol{d})\\ &\quad+\big(M_{10}-M_{11}\big)(\boldsymbol{d}+\boldsymbol{e}_x)+\big(M_{01}-M_{11}\big)(\boldsymbol{d}-\boldsymbol{e}_x), \end{aligned} \tag{10}$$

with $M = M^{[g]}$. For $d_x = 0$ the last term is to be read as

$$\big(M_{01} - M_{11}\big)(-1, d_y) = \big(M_{10} - M_{11}\big)(1, d_y). \tag{11}$$

Furthermore $\Phi(\boldsymbol{d}) = (ab)^2 M_{00}^{[g]}(\boldsymbol{d})$, and A^{yy} is given by (10) with N and $\boldsymbol{e}_y$ in place of M and $\boldsymbol{e}_x$.

Proof. Split both rooftops into their wedges, as in Definition (2.2). The reference rooftop $\boldsymbol{\Psi}_{\mathbf{0}}^x$ has the rising wedge ξ on $\Omega_{\mathbf{0}}$ and the falling wedge $1-\xi$ on $\Omega_{\boldsymbol{e}_x}$. The rooftop $\boldsymbol{\Psi}_{\boldsymbol{d}}^x$ has the rising wedge ξ' on $\Omega_{\boldsymbol{d}}$ and the falling wedge $1-\xi'$ on $\Omega_{\boldsymbol{d}+\boldsymbol{e}_x}$. Both are x-directed, so the dot product is the product of the profiles. The four wedge pairs of Figure 2 give

pair	cells	offset	weight	moment
1, rising–rising	$\Omega_{\mathbf{0}}$, $\Omega_{\boldsymbol{d}}$	$\boldsymbol{d}$	$\xi\xi'$	M_{11}
2, falling–falling	$\Omega_{\boldsymbol{e}_x}$, $\Omega_{\boldsymbol{d}+\boldsymbol{e}_x}$	$\boldsymbol{d}$	$(1-\xi)(1-\xi')$	$M_{00} - M_{10} - M_{01} + M_{11}$
3, rising–falling	$\Omega_{\mathbf{0}}$, $\Omega_{\boldsymbol{d}+\boldsymbol{e}_x}$	$\boldsymbol{d}+\boldsymbol{e}_x$	$\xi(1-\xi')$	$M_{10} - M_{11}$
4, falling–rising	$\Omega_{\boldsymbol{e}_x}$, $\Omega_{\boldsymbol{d}}$	$\boldsymbol{d}-\boldsymbol{e}_x$	$(1-\xi)\xi'$	$M_{01} - M_{11}$

with the moments taken at the offset of the pair (for pair 4 after the translation by $-\boldsymbol{e}_x$). The normalization $(ab)^{-2}$ in Definition (2.3) accounts for the factor $(ab)^2$. For $d_x = 0$ the offset $\boldsymbol{d} - \boldsymbol{e}_x = (-1, d_y)$ is negative. Lemma (2.1)(ii) gives

$$\begin{aligned}(M_{01} - M_{11})(-1, d_y) &= (M_{00} - M_{01})(1, d_y) - (M_{00} - M_{10} - M_{01} + M_{11})(1, d_y) \\ &= (M_{10} - M_{11})(1, d_y).\end{aligned} \tag{12}$$

The statement on Φ is the definition with $\alpha = \beta = 0$. The y-direction is identical. □

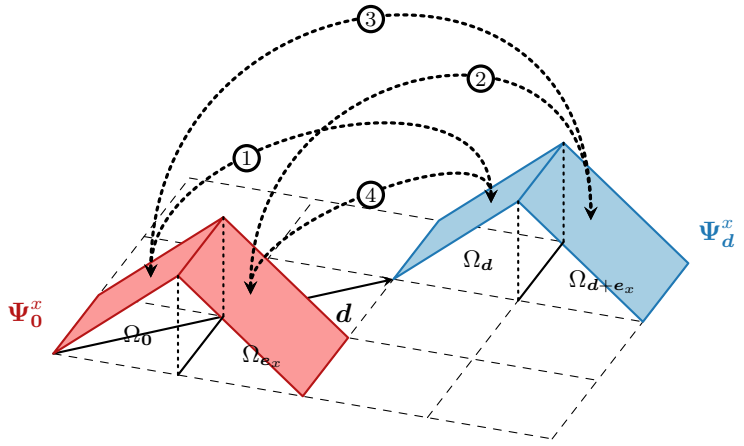


Figure 2: The four wedge pairs of the rooftops $\boldsymbol{\Psi}_{\mathbf{0}}^x$ and $\boldsymbol{\Psi}_{\boldsymbol{d}}^x$ in Proposition (2.1), drawn for $\boldsymbol{d} = (2, 2)$, numbered as in its proof. Every pair is a pair of cells at one offset, weighted by a product of two linear profiles.

The generators of all three Toeplitz blocks need the seven moments on the offsets $d_x \geq -1$, $d_y \geq 0$, and by Lemma (2.1) on $d_x, d_y \geq 0$ only.

3 Singularity extraction on the touching cells

Proposition 3.1 (Non-smooth terms of the Green's function). For $R \geq 0$,

$$g(R) = \frac{\mathrm{e}^{-\mathrm{i}kR}}{R} = \sum_{n=0}^{\infty} \frac{(-\mathrm{i}k)^n}{n!} R^{n-1} = \sum_{\ell=0}^{\infty} \frac{(-1)^\ell k^{2\ell}}{(2\ell)!} R^{2\ell-1} \; - \; \mathrm{i} \sum_{\ell=0}^{\infty} \frac{(-1)^\ell k^{2\ell+1}}{(2\ell+1)!} R^{2\ell}. \quad (13)$$

The series converge absolutely and locally uniformly for $R > 0$; the second one is an entire function of R^2. As functions of $\boldsymbol{z} = \boldsymbol{r} - \boldsymbol{r}' \in \mathbb{R}^2$ the even powers $R^{2\ell} = \|\boldsymbol{z}\|^{2\ell}$ are polynomials, while the odd powers $R^{2\ell-1}$ are not smooth at $\boldsymbol{z} = \boldsymbol{0}$.

Proof. Insert the exponential series and separate the even and odd n; this is Proposition 2.2 in Section 2.6 of [1].

□

Definition 3.1 (Extraction of order N). For $N \in \mathbb{N}$ let

$$S_N(R) = \sum_{\ell=0}^{N-1} \frac{(-1)^\ell k^{2\ell}}{(2\ell)!} R^{2\ell-1} = \frac{1}{R} - \frac{k^2}{2} R + \frac{k^4}{24} R^3 - \dots, \qquad g_N = g - S_N. \quad (14)$$

Table 1 lists the orders: $N = 1$ is the classical extraction of the static singularity [2], $N = 2$ the two-term extraction of [1] (Entry 2.1 in Section 2.6), there written as

$$g = g_{\mathrm{s}} + \frac{1}{4\pi} R^{-1} - \frac{k^2}{8\pi} R, \quad (15)$$

and $N = 3$ the one of this note. For every $N \geq 1$, g_N extends continuously to $R = 0$ with $g_N(0) = -\mathrm{i}k$.

Lemma 3.1 (Regularity of the remainder). For $N \geq 1$ the function $\boldsymbol{z} \mapsto g_N(\|\boldsymbol{z}\|)$ is real-analytic on $\mathbb{R}^2 \setminus \{\boldsymbol{0}\}$ and belongs to $C^{2N-2,1}$ on every bounded set: its derivatives up to the order $2N-2$ are continuous, and those of the order $2N-2$ are Lipschitz. It is not of class C^{2N-1} unless $k = 0$.

Proof. By (13),

$$g_N(R) = E(R^2) + R^{2N-1} F(R^2), \quad (16)$$

with the entire functions

$$E(s) = -\mathrm{i} \sum_{\ell \geq 0} \frac{(-1)^\ell k^{2\ell+1}}{(2\ell+1)!} s^\ell, \qquad F(s) = \sum_{\ell \geq N} \frac{(-1)^\ell k^{2\ell}}{(2\ell)!} s^{\ell-N}. \quad (17)$$

The function $E(\|\boldsymbol{z}\|^2)$ is real-analytic on $\mathbb{R}^2$.

The function $\|\boldsymbol{z}\|^{2N-1} = \left(\|\boldsymbol{z}\|^2\right)^{N-1} \|\boldsymbol{z}\|$ is smooth away from the origin and positively homogeneous of degree $2N-1$. Hence its derivatives of order $m \leq 2N-2$ are homogeneous of degree $2N-1-m \geq 1$. They are therefore continuous at the origin (with the value zero), and those of order $2N-2$, being homogeneous of degree one, are Lipschitz. The derivatives of order $2N-1$ are homogeneous of degree zero and not constant, so they are bounded but discontinuous at the origin.

Multiplication by the analytic factor $F(\|\boldsymbol{z}\|^2)$ preserves the class $C^{2N-2,1}$ on bounded sets. Since $F(0) = (-1)^N k^{2N}/(2N)! \neq 0$ for $k \neq 0$, the term $F(0)\,\|\boldsymbol{z}\|^{2N-1}$ is present and the class is sharp.

□

On a touching cell pair the integrand of a moment meets $\boldsymbol{z} = \mathbf{0}$, in the interior of the four-dimensional domain for the coincident cell and on its boundary for a shared edge or vertex. A tensor GL rule integrates polynomials exactly, so its error on such a pair is governed by the regularity of Lemma (3.1) (Table 1).

Table 1: The extraction orders and the regularity of the remainder on a touching cell pair.

N	extracted S_N	used in	remainder g_N
0	—		unbounded
1	$1/R$	[2]	Lipschitz
2	$1/R - k^2R/2$	[1]	$C^{2,1}$
3	$1/R - k^2R/2 + k^4R^3/24$	this note	$C^{4,1}$

The error then decreases algebraically in the order q, and the faster, the larger N. The lemma is not a quadrature error estimate; for that theory see [3]. On the cell pairs that do not touch, g is real-analytic on a neighbourhood of the domain, and the GL error decreases exponentially.

Corollary 3.1 (Exact decomposition of the touching moments). For every $N \geq 1$, $\boldsymbol{d} \in \mathcal{T}$ and $\alpha, \beta \in \{0, 1\}$,

$$M^{[g]}_{\alpha\beta}(\boldsymbol{d}) = M^{[g_N]}_{\alpha\beta}(\boldsymbol{d}) + \sum_{\ell=0}^{N-1} \frac{(-1)^\ell k^{2\ell}}{(2\ell)!}\, M^{(2\ell-1)}_{\alpha\beta}(\boldsymbol{d}), \tag{18}$$

and likewise for the N-moments. The first term is evaluated with GL; the static moments $M^{(p)}_{\alpha\beta}(\boldsymbol{d})$, $p \in \{-1, 1, 3\}$, are given in closed form by Theorem (6.1).

Proof. The moments are linear in K, and every term is absolutely integrable: the kernels $R^{2\ell-1}$ with $\ell \geq 0$ have $2\ell - 1 \geq -1 > -2$, and g_N is bounded. □

What has to be computed. By Lemma (2.1), the static moments of the four offsets

$$\boldsymbol{d} \in \{(0,0),\ (1,0),\ (0,1),\ (1,1)\} \tag{19}$$

determine all nine touching offsets. These are the coincident cell, the two kinds of shared edge and the shared vertex. For $N = 3$ we need, for each of them, the four x-moments $M_{00}, M_{10}, M_{01}, M_{11}$ and the three y-moments, with $p \in \{-1, 1, 3\}$. The y-moments follow from the x-moments by Lemma (2.1)(iii).

4 Reduction to two dimensions

The integrand of a moment depends on the four coordinates only through the two differences $x - x'$ and $y - y'$, and the weights are polynomials. Integrating out the "position along the diagonal" in each direction leaves a two-dimensional integral with piecewise polynomial weights. The coincident-cell transformation of [1] (Proposition 2.3 in Section 2.6, regions I–IV), which follows [4], is the special case $\boldsymbol{d} = \mathbf{0}$.

Lemma 4.1 (One-dimensional correlation). Let $\varphi, \psi \in L^\infty(0,1)$, $\delta \in \mathbb{R}$, and let $h : \mathbb{R} \to \mathbb{C}$ be locally integrable. Then

$$\int_0^1\!\!\int_0^1 \varphi(\xi)\,\psi(\tau)\,h(\xi-\tau-\delta)\,\mathrm{d}\tau\,\mathrm{d}\xi = \int_{-1}^{1} W_{\varphi\psi}(\nu)\,h(\nu-\delta)\,\mathrm{d}\nu,$$
$$W_{\varphi\psi}(\nu) = \int_{\max(0,\nu)}^{\min(1,1+\nu)} \varphi(\xi)\,\psi(\xi-\nu)\,\mathrm{d}\xi, \tag{20}$$

and both integrals converge absolutely.

Proof. The linear map $(\xi,\tau) \mapsto (\xi,\nu)$, $\nu = \xi - \tau$, has determinant one. It maps the unit square onto the parallelogram (Figure 3)

$$\{(\xi,\nu) :\ -1 \le \nu \le 1,\ \max(0,\nu) \le \xi \le \min(1,1+\nu)\}, \tag{21}$$

because $0 \le \tau = \xi - \nu \le 1$ is equivalent to $\nu \le \xi \le 1+\nu$.

By Tonelli's theorem applied to $|\varphi|\,|\psi|\,|h|$, the left-hand side is absolutely convergent, since

$$\int_0^1\!\!\int_0^1 |\varphi(\xi)\,\psi(\tau)\,h(\xi-\tau-\delta)|\,\mathrm{d}\tau\,\mathrm{d}\xi \le \|\varphi\|_\infty\,\|\psi\|_\infty \int_{-1}^{1} (1-|\nu|)\,|h(\nu-\delta)|\,\mathrm{d}\nu < \infty. \tag{22}$$

Fubini's theorem then allows the integration in ξ first, for fixed ν, which gives $W_{\varphi\psi}(\nu)$. □

Corollary 4.1 (Weights of the rooftop moments). For $\varphi(\xi) = \xi^\alpha$ and $\psi(\tau) = \tau^\beta$, $\alpha, \beta \in \{0,1\}$, write $W_{\alpha\beta} = W_{\varphi\psi}$. On $I_- = [-1,0]$ and $I_+ = [0,1]$ these are the polynomials

	$\nu \in I_-$	$\nu \in I_+$
W_{00}	$1+\nu$	$1-\nu$
W_{10}	$\tfrac{1}{2}(1+\nu)^2$	$\tfrac{1}{2}(1-\nu^2)$
W_{01}	$\tfrac{1}{2}(1-\nu^2)$	$\tfrac{1}{2}(1-\nu)^2$
W_{11}	$\tfrac{1}{6}(1+\nu)^2(2-\nu)$	$\tfrac{1}{6}(2-3\nu+\nu^3)$

(23)

They are continuous on $[-1,1]$, W_{00} and W_{11} are even, and $W_{01}(\nu) = W_{10}(-\nu)$.

Proof. For $\nu \in I_+$ the limits in (20) are $\nu \le \xi \le 1$:

$$W_{00} = \int_\nu^1 \mathrm{d}\xi = 1-\nu, \qquad W_{10} = \int_\nu^1 \xi\,\mathrm{d}\xi = \frac{1}{2}(1-\nu^2),$$
$$W_{01} = \int_\nu^1 (\xi-\nu)\,\mathrm{d}\xi = \frac{1}{2}(1-\nu)^2, \quad W_{11} = \int_\nu^1 \xi(\xi-\nu)\,\mathrm{d}\xi = \frac{1}{6}\big(2-3\nu+\nu^3\big). \tag{24}$$

For $\nu \in I_-$ they are $0 \le \xi \le 1+\nu$:

$$\begin{aligned} W_{00} &= \int_0^{1+\nu} \mathrm{d}\xi = 1+\nu, & W_{10} &= \int_0^{1+\nu} \xi \,\mathrm{d}\xi = \frac{1}{2}(1+\nu)^2, \\ W_{01} &= \int_0^{1+\nu} (\xi-\nu)\,\mathrm{d}\xi = \frac{1}{2}(1-\nu^2), & W_{11} &= \int_0^{1+\nu} \xi(\xi-\nu)\,\mathrm{d}\xi = \frac{1}{6}(1+\nu)^2(2-\nu). \end{aligned} \tag{25}$$

At $\nu = 0$ both sides give the values 1, 1/2, 1/2 and 1/3, so the weights are continuous. For the symmetries, the substitution $s = \xi + \nu$ in (20) gives $W_{\varphi\psi}(-\nu) = W_{\psi\varphi}(\nu)$. Hence $W_{01}(\nu) = W_{10}(-\nu)$, and W_{00}, W_{11} are even; for W_{11} compare

$$\frac{1}{6}(1-\nu)^2(2+\nu) = \frac{1}{6}\big(2 - 3\nu + \nu^3\big). \tag{26}$$

□

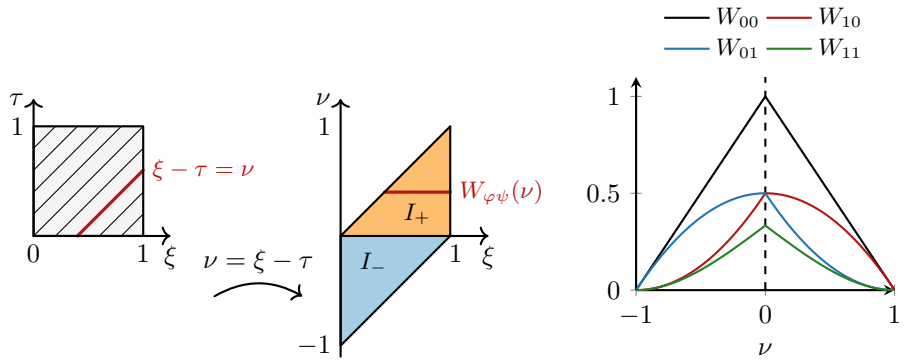


Figure 3: Left: the substitution $\nu = \xi - \tau$ of Lemma (4.1) maps the unit square onto a parallelogram; the line $\xi - \tau = \nu$ (red) becomes the horizontal segment over which $W_{\varphi\psi}(\nu)$ integrates. The two triangles $\nu \in I_-$ and $\nu \in I_+$ have different limits in ξ, which is why the weights change their polynomial form at $\nu = 0$. Right: the weights (23).

Theorem 4.1 (Two-dimensional representation). Let $p > -2$, $\boldsymbol{d} = (d_x, d_y) \in \mathbb{Z}^2$ and $\alpha, \beta \in \{0, 1\}$, and put

$$\rho_{\boldsymbol{d}}(\nu, \eta) = \sqrt{a^2(\nu - d_x)^2 + b^2(\eta - d_y)^2}. \tag{27}$$

Then

$$\begin{aligned} M^{(p)}_{\alpha\beta}(\boldsymbol{d}) &= \int_{-1}^{1}\int_{-1}^{1} W_{\alpha\beta}(\nu)\, W_{00}(\eta)\, \rho_{\boldsymbol{d}}(\nu,\eta)^p \,\mathrm{d}\eta\,\mathrm{d}\nu, \\ N^{(p)}_{\alpha\beta}(\boldsymbol{d}) &= \int_{-1}^{1}\int_{-1}^{1} W_{00}(\nu)\, W_{\alpha\beta}(\eta)\, \rho_{\boldsymbol{d}}(\nu,\eta)^p \,\mathrm{d}\eta\,\mathrm{d}\nu. \end{aligned} \tag{28}$$

The same holds for a general admissible kernel K, with $K(\rho_{\boldsymbol{d}})$ in place of $\rho^p_{\boldsymbol{d}}$.

Proof. Parametrize $\boldsymbol{r} = (a\xi_1, b\eta_1) \in \Omega_{\boldsymbol{0}}$ and $\boldsymbol{r}' = \big(a(d_x + \xi_2),\, b(d_y + \eta_2)\big) \in \Omega_{\boldsymbol{d}}$ with $\xi_1, \eta_1, \xi_2, \eta_2 \in [0, 1]$. Then $\mathrm{d}A'\,\mathrm{d}A = (ab)^2\,\mathrm{d}\eta_2\,\mathrm{d}\xi_2\,\mathrm{d}\eta_1\,\mathrm{d}\xi_1$, which cancels the normaliza-

tion of Definition (2.3), $\xi(\boldsymbol{r}) = \xi_1$, $\xi(\boldsymbol{r}') = \xi_2$, and

$$\|\boldsymbol{r} - \boldsymbol{r}'\| = \sqrt{a^2(\xi_1 - \xi_2 - d_x)^2 + b^2(\eta_1 - \eta_2 - d_y)^2}. \tag{29}$$

The moment is therefore the integral over the unit cube $[0,1]^4$

$$M^{(p)}_{\alpha\beta}(\boldsymbol{d}) = \int_0^1\int_0^1\int_0^1\int_0^1 \xi_1^\alpha\, \xi_2^\beta \left(a^2(\xi_1-\xi_2-d_x)^2 + b^2(\eta_1-\eta_2-d_y)^2\right)^{p/2} \mathrm{d}\eta_2\, \mathrm{d}\xi_2\, \mathrm{d}\eta_1\, \mathrm{d}\xi_1. \tag{30}$$

The integrand depends on the coordinates of the two directions only through the differences $\xi_1 - \xi_2$ and $\eta_1 - \eta_2$, and the weight depends on the x-coordinates only. The proof integrates out one difference after the other with Lemma (4.1), first in x and then in y.

First, the integral converges absolutely. The function $\boldsymbol{z} \mapsto \|\boldsymbol{z}\|^p$ is locally integrable on $\mathbb{R}^2$ for $p > -2$ (polar coordinates). Lemma (4.1), applied in each direction to the nonnegative kernel by Tonelli's theorem, gives

$$\int\!\!\int\!\!\int\!\!\int |\cdot| = \int_{-1}^1\int_{-1}^1 (1-|\nu|)(1-|\eta|)\, \rho_{\boldsymbol{d}}^p \le \int_{-1}^1\int_{-1}^1 \rho_{\boldsymbol{d}}^p < \infty. \tag{31}$$

Fubini's theorem therefore allows the four integrations of (30) in any order.

Second, the x-direction. Fix η_1 and η_2, and write $c = b(\eta_1 - \eta_2 - d_y)$ for the separation of the two points in y. The integrals over ξ_1 and ξ_2 in (30) are then

$$\int_0^1\int_0^1 \xi_1^\alpha\, \xi_2^\beta \left(a^2(\xi_1-\xi_2-d_x)^2 + c^2\right)^{p/2} \mathrm{d}\xi_2\, \mathrm{d}\xi_1. \tag{32}$$

This is the left-hand side of (20) with $\varphi(\xi) = \xi^\alpha$, $\psi(\tau) = \tau^\beta$, $\delta = d_x$ and the function of one real variable

$$h(s) = \left(a^2 s^2 + c^2\right)^{p/2}, \qquad s \in \mathbb{R}. \tag{33}$$

In the lemma h is evaluated at $s = \xi_1 - \xi_2 - d_x$. Then $a^2 s^2 = (x - x')^2$ is the squared separation of the two points in x, and $a^2 s^2 + c^2 = \|\boldsymbol{r} - \boldsymbol{r}'\|^2$. For $c \neq 0$ the function h is continuous, hence locally integrable. The pairs with $c = 0$, that is $\eta_1 - \eta_2 = d_y$, form a line in the (η_1, η_2)-square and do not contribute to the outer integral. The lemma, with the weight $W_{\varphi\psi} = W_{\alpha\beta}$ of Corollary (4.1), gives

$$\int_0^1\int_0^1 \xi_1^\alpha\, \xi_2^\beta \left(a^2(\xi_1-\xi_2-d_x)^2 + c^2\right)^{p/2} \mathrm{d}\xi_2\, \mathrm{d}\xi_1 = \int_{-1}^1 W_{\alpha\beta}(\nu) \left(a^2(\nu-d_x)^2 + c^2\right)^{p/2} \mathrm{d}\nu. \tag{34}$$

The two x-coordinates are replaced by their difference ν, and the weights ξ_1^α, ξ_2^β by the single weight $W_{\alpha\beta}(\nu)$. With $c = b(\eta_1 - \eta_2 - d_y)$ inserted back,

$$M^{(p)}_{\alpha\beta}(\boldsymbol{d}) = \int_{-1}^1 W_{\alpha\beta}(\nu) \left(\int_0^1\int_0^1 \left(a^2(\nu-d_x)^2 + b^2(\eta_1-\eta_2-d_y)^2\right)^{p/2} \mathrm{d}\eta_2\, \mathrm{d}\eta_1 \right) \mathrm{d}\nu, \tag{35}$$

where the ν-integration has been moved outside by Fubini's theorem.

Third, the y-direction. Fix $\nu \neq d_x$; the single value $\nu = d_x$ again does not contribute. The inner integral in the last display is the left-hand side of (20) once more, now in the variables (η_1, η_2) with $\varphi = \psi = 1$, $\delta = d_y$ and

$$h(s) = \left(a^2(\nu - d_x)^2 + b^2 s^2\right)^{p/2}, \qquad s \in \mathbb{R}, \tag{36}$$

where s now stands for $\eta_1 - \eta_2 - d_y$, so that $b^2 s^2 = (y - y')^2$ is the squared separation of the two points in y. For the weights 1 and 1 the weight of the lemma is W_{00}, so

$$\int_0^1 \int_0^1 \Big(a^2(\nu - d_x)^2 + b^2(\eta_1 - \eta_2 - d_y)^2 \Big)^{p/2} \, \mathrm{d}\eta_2 \, \mathrm{d}\eta_1 = \int_{-1}^{1} W_{00}(\eta) \, \rho_{\boldsymbol{d}}(\nu, \eta)^p \, \mathrm{d}\eta. \tag{37}$$

Inserting this into the previous display gives the first line of (28).

Finally, the N-moments carry the weights $\eta_1^\alpha \eta_2^\beta$ in the y-coordinates instead. The same three steps then give $W_{00}(\nu)$ in x and $W_{\alpha\beta}(\eta)$ in y, which is the second line of (28). Nothing in the argument uses the form of the power ρ^p beyond its local integrability, so a general admissible kernel K gives the same result with $K(\rho_{\boldsymbol{d}})$ in place of $\rho_{\boldsymbol{d}}^p$. □

Remark. For $\boldsymbol{d} \in \mathcal{T}$ the singular point of $\rho_{\boldsymbol{d}}^p$, $p < 0$, is $(\nu, \eta) = (d_x, d_y)$. It is the centre, a midpoint of an edge or a corner of $[-1, 1]^2$ (Figure 6). In every case it lies on the lines $\nu \in \{-1, 0, 1\}$ and $\eta \in \{-1, 0, 1\}$, where the weights change their polynomial form. This is the structural fact behind the closed forms of Section 6.

5 Mixed antiderivatives

The antiderivatives below are functions of two lengths, which are written as a and b like in the thesis appendix. This does not clash with the cell sizes: the touching moments evaluate them only at the points (a, b), $(2a, b)$, $(a, 2b)$ and $(2a, 2b)$ (Section 6).

Definition 5.1 (Mixed antiderivatives). Let $Q = [0, \infty)^2$ be the closed first quadrant and Q° its interior. For $p > -2$, $m, n \in \mathbb{N}_0$ and $(a, b) \in Q$ put

$$G_{mn}^{(p)}(a, b) = \int_0^a \int_0^b u^m v^n \big(u^2 + v^2\big)^{p/2} \, \mathrm{d}v \, \mathrm{d}u. \tag{38}$$

For a function F on Q and a rectangle $A \times B \subset Q$ with $A = [a_0, a_1]$ and $B = [b_0, b_1]$, the corner difference of F over $A \times B$ is

$$\Delta_{A \times B} F = F(a_1, b_1) - F(a_0, b_1) - F(a_1, b_0) + F(a_0, b_0). \tag{39}$$

Proposition 5.1 (Properties of G). Let $p > -2$ and $m, n \in \mathbb{N}_0$.

(i) $G_{mn}^{(p)}$ is continuous on Q and vanishes on both axes: $G_{mn}^{(p)}(a, 0) = G_{mn}^{(p)}(0, b) = 0$.

(ii) For every rectangle $[a_0, a_1] \times [b_0, b_1] \subset Q$,

$$\int_{a_0}^{a_1} \int_{b_0}^{b_1} u^m v^n \big(u^2 + v^2\big)^{p/2} \, \mathrm{d}v \, \mathrm{d}u = \Delta_{[a_0, a_1] \times [b_0, b_1]} G_{mn}^{(p)}. \tag{40}$$

(iii) $\partial_a \partial_b G_{mn}^{(p)} = a^m b^n \big(a^2 + b^2\big)^{p/2}$ on Q°, and $G_{mn}^{(p)}$ is the only function in $C(Q) \cap C^2(Q^\circ)$ with this property that vanishes on both axes.

(iv) $G_{mn}^{(p)}(\lambda a, \lambda b) = \lambda^{m+n+p+2} \, G_{mn}^{(p)}(a, b)$ for $\lambda > 0$.

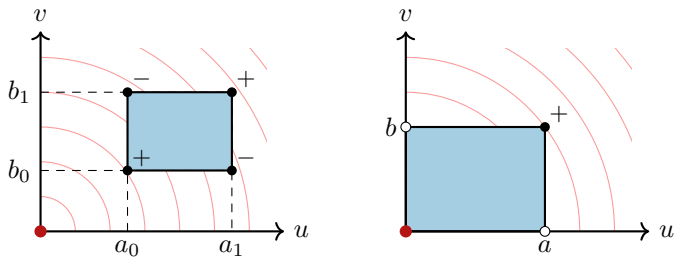


Figure 4: Left: the corner difference (40). The integral over the shaded rectangle is the alternating sum of $G^{(p)}_{mn}$ at its corners, each value being the integral over $[0,a]\times[0,b]$ (dashed). The circles are level lines of $\sqrt{u^2+v^2}$ around the singular point (red). Right: a piece that touches the singular point. Three of its corners lie on the axes, where $G^{(p)}_{mn}$ vanishes, and only $G^{(p)}_{mn}(a,b)$ remains.

Proof. (i) The integrand is locally integrable on Q because $p>-2$; continuity follows by dominated convergence, and the integral over a rectangle with a side of length zero is zero.

(ii) This is inclusion–exclusion (Figure 4): the rectangle $[a_0,a_1]\times[b_0,b_1]$ is $[0,a_1]\times[0,b_1]$ minus $[0,a_0]\times[0,b_1]$ and $[0,a_1]\times[0,b_0]$, plus their overlap $[0,a_0]\times[0,b_0]$.

(iii) The derivative follows from the fundamental theorem of calculus in each variable. If D is the difference of two such functions, $\partial_a\partial_b D=0$ gives $D(a,b)=e(a)+c(b)$ on Q°; letting $b\to0^+$ and $a\to0^+$ with $D=0$ on the axes shows that e and c are constants with $e+c=D(0,0)=0$.

(iv) Substitute $u=\lambda u'$ and $v=\lambda v'$ in (38).

□

Lemma 5.1 (One-dimensional integrals). Let $a\ge0$, $b>0$, $m\in\mathbb{N}_0$ and $q\in\{-1,1,3,5\}$, and let $R=\sqrt{a^2+b^2}$. The integrals

$$K^{(q)}_m(a)=\int_0^a u^m\bigl(u^2+b^2\bigr)^{q/2}\,\mathrm{d}u \tag{41}$$

are

$$K^{(q)}_0=\begin{cases}\log\left(\dfrac{R+a}{b}\right), & q=-1,\\[2mm] \dfrac{1}{2}\Bigl(aR+b^2\log\left(\dfrac{R+a}{b}\right)\Bigr), & q=1,\\[2mm] \dfrac{1}{8}\Bigl(2aR^3+3b^2aR+3b^4\log\left(\dfrac{R+a}{b}\right)\Bigr), & q=3,\\[2mm] \dfrac{1}{48}\Bigl(8aR^5+10b^2aR^3+15b^4aR+15b^6\log\left(\dfrac{R+a}{b}\right)\Bigr), & q=5,\end{cases} \tag{42}$$

$$K^{(q)}_1=\frac{R^{q+2}-b^{q+2}}{q+2}, \tag{43}$$

$$K^{(q)}_m=\frac{a^{m-1}R^{q+2}-(m-1)b^2K^{(q)}_{m-2}}{m+q+1},\qquad m\ge2, \tag{44}$$

and, for the logarithmic terms produced by the inner integrations,

$$\int_0^a u^m \log\left(\frac{\sqrt{u^2+b^2}+b}{u}\right)\mathrm{d}u = \frac{1}{m+1}\Big(a^{m+1}\log\left(\frac{R+b}{a}\right) + b\,K_m^{(-1)}(a)\Big). \tag{45}$$

The same formulas with a and b exchanged give the inner integrals $\int_0^b v^n(u^2+v^2)^{p/2}\,\mathrm{d}v$.

Proof. The base cases follow from

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}u}\log\left(\frac{\sqrt{u^2+b^2}+u}{b}\right) &= \big(u^2+b^2\big)^{-1/2},\\
\frac{\mathrm{d}}{\mathrm{d}u}\big(u^2+b^2\big)^{(q+2)/2} &= (q+2)\,u\,\big(u^2+b^2\big)^{q/2},\\
\frac{\mathrm{d}}{\mathrm{d}u}\Big(u\big(u^2+b^2\big)^{q/2}\Big) &= (q+1)\big(u^2+b^2\big)^{q/2} - qb^2\big(u^2+b^2\big)^{(q-2)/2};
\end{aligned} \tag{46}$$

the first one gives $K_0^{(-1)}$ and the second one (43). The last one gives the recursion

$$K_0^{(q)} = \frac{aR^q + qb^2K_0^{(q-2)}}{q+1}, \tag{47}$$

which produces (42) from $K_0^{(-1)}$.

For the recursion with $m \geq 2$, integrate from 0 to a

$$\frac{\mathrm{d}}{\mathrm{d}u}\big(u^{m-1}(u^2+b^2)^{(q+2)/2}\big) = (m-1)b^2u^{m-2}(u^2+b^2)^{q/2} + (m+q+1)u^m(u^2+b^2)^{q/2}. \tag{48}$$

The logarithmic terms follow by integration by parts with

$$\frac{\mathrm{d}}{\mathrm{d}u}\log\left(\frac{\sqrt{u^2+b^2}+b}{u}\right) = -\frac{b}{u\sqrt{u^2+b^2}}; \tag{49}$$

the boundary term $u^{m+1}\log\big((\sqrt{u^2+b^2}+b)/u\big)$ vanishes as $u \to 0^+$.

□

Proposition 5.2 (Closed forms). For $p \in \{-1,1,3\}$, $m \in \{0,1,2,3\}$, $n \in \{0,1\}$ and $(a,b) \in Q^\circ$,

$$\begin{aligned}
G_{mn}^{(p)}(a,b) = \frac{1}{D}\Bigg(&P_0 + P_1\sqrt{a^2+b^2} + P_2\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\\
&+ P_3\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right)\Bigg),
\end{aligned} \tag{50}$$

with an integer D and homogeneous integer polynomials $P_0,\dots,P_3$ in a and b, listed in Appendix A. These are all the antiderivatives the moments need; any other one follows from the symmetry $G_{nm}^{(p)}(b,a) = G_{mn}^{(p)}(a,b)$.

Proof. The formulas are obtained by integrating first in v and then in u with Lemma (5.1); every step produces only terms of the form (50).

□

Example. For $p = -1$, (45) gives directly

$$\begin{aligned} G_{00}^{(-1)}(a,b) &= \int_0^a \log\left(\frac{\sqrt{u^2+b^2}+b}{u}\right) \mathrm{d}u \\ &= a\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) + b\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right), \end{aligned} \tag{51}$$

the classical potential integral of a uniformly charged rectangle at its corner.

6 Closed forms for the touching cells

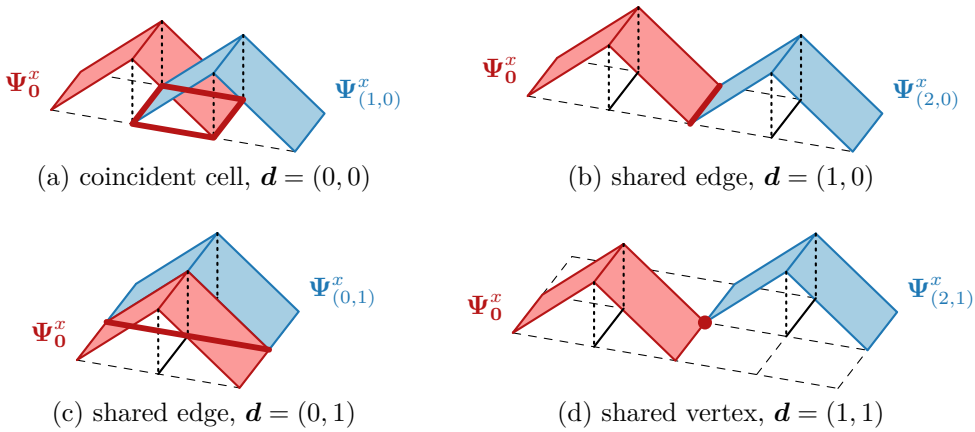
Within the note, this section combines three earlier results: the two-dimensional form of a moment (Theorem (4.1)), the weights that are polynomials on I_- and I_+ (Corollary (4.1)), and the corner difference with the closed-form antiderivatives $G_{mn}^{(p)}$ (Proposition (5.1) and Appendix A). In the literature, closed forms of static potential integrals are classical. Wilton et al. [5] give them for uniform and linear sources on polygons, a single integration over the source domain, and Tzoulis and Eibert [2] review this line of work. Lopez-Pena and Mosig [6] evaluate the full fourfold static integrals of uniform sources over rectangular cells, and Bao et al. [4] and the thesis [1] do so for rooftop functions on the coincident cell. This section extends the last two to the linear weights of the rooftops on all nine touching cells, the edge and vertex neighbours included, and from the kernel R^{-1} to the kernels R and R^3 of the three-term extraction.

Theorem (4.1) turns a moment into the two-dimensional integral (28) over $[-1,1]^2$, with the weights $W_{\alpha\beta}(\nu)$, $W_{00}(\eta)$ and the distance $\rho_{\boldsymbol{d}}(\nu,\eta)$. For a touching offset $\boldsymbol{d} = (d_x, d_y) \in \{0,1\}^2$ the distance vanishes at the singular point $(\nu,\eta) = (d_x, d_y)$, and this is the only point where the integrand is not smooth. The plan of this section is to cut $[-1,1]^2$ into pieces on which the weights are polynomials and to move the singular point of every piece to the origin. Each piece then becomes a rectangle in the quadrant Q of Section 5, with polynomial weights in front of $(u^2+v^2)^{p/2}$, and Proposition (5.1)(ii) integrates it exactly with the antiderivatives $G_{mn}^{(p)}$.

Figure 5 shows where these configurations come from. A rooftop consists of two wedges, so a pair of rooftops gives four wedge pairs (Proposition (2.1)), and each wedge pair is a pair of cells. Two rooftops in a row that overlap by one cell have a wedge pair on the same cell, the coincident configuration. With one cell between them, the two inner wedges share an edge parallel to y. Two rooftops side by side in y have wedge pairs that share an edge parallel to x, and two rooftops placed diagonally have a wedge pair that meets in a single vertex. In all four cases the two cells of the pair have the points of the red set in common, where $R = 0$.

The cut is the one of Corollary (4.1): in each direction the interval $[-1,1]$ splits into $I_- = [-1,0]$ and $I_+ = [0,1]$, and on each of them the weights are polynomials. This gives the four pieces $I_s \times I_t$, $s,t \in \{-,+\}$, of Figure 6.

Moving the singular point to the origin is what makes one table of closed forms enough. First, at the origin the kernel is $(u^2+v^2)^{p/2}$ without any shift, so every piece of every configuration is integrated by the same functions $G_{mn}^{(p)}$ of Definition (5.1), which depend only on the end point (a,b) of the rectangle. A singular point at some other position (u_0,v_0) would need a separate family of antiderivatives for every shift. Second, on a piece with $u,v \geq 0$ the absolute values $|x-x'|$ and $|y-y'|$ in the distance and the kinks of the weights at $\nu = d_x$ and $\eta = d_y$ disappear, so the integrand is a sum of terms $u^m v^n (u^2+v^2)^{p/2}$, the integrand of (38). Third, the singular point is a corner of every piece that contains it, and $G_{mn}^{(p)}$ vanishes on both axes (Proposition (5.1)(i)). The corner



(a) coincident cell, $\boldsymbol{d} = (0, 0)$

(b) shared edge, $\boldsymbol{d} = (1, 0)$

(c) shared edge, $\boldsymbol{d} = (0, 1)$

(d) shared vertex, $\boldsymbol{d} = (1, 1)$

Figure 5: Two x-directed rooftops, $\boldsymbol{\Psi}^x_{\mathbf{0}}$ (red) and a second one (blue), placed so that two of their wedges lie on touching cells; the set that the two cells have in common is drawn in red on the floor. (a) Rooftop offset $(1, 0)$: the falling wedge of $\boldsymbol{\Psi}^x_{\mathbf{0}}$ and the rising wedge of $\boldsymbol{\Psi}^x_{(1,0)}$ lie on the same cell. (b) Rooftop offset $(2, 0)$: the wedges on $\Omega_{(1,0)}$ and $\Omega_{(2,0)}$ share an edge parallel to y. (c) Rooftop offset $(0, 1)$: the wedges side by side share edges parallel to x. (d) Rooftop offset $(2, 1)$: the wedges on $\Omega_{(1,0)}$ and $\Omega_{(2,1)}$ share only a vertex.

sum (40) therefore never evaluates anything at $R = 0$, and the singularity is integrated exactly, without a limit. Placing the singularity at a vertex is also the starting point of the Duffy transformation [7], which uses it to make the integrand smooth for quadrature instead of integrating it in closed form. The next lemma moves the singular point to the origin, one direction at a time.

Lemma 6.1 (The polynomial pieces). Let $d \in \{0, 1\}$ be the offset in the x-direction, so that the singular value of ν is $\nu = d$. On each interval I_s, $s \in \{-, +\}$, introduce the distance from the singular value, measured in units of length,

$$u = a\,|\nu - d|\,. \tag{52}$$

On I_s the difference $\nu - d$ does not change sign, so $|\nu - d| = \sigma(\nu - d)$ with a fixed sign $\sigma \in \{-1, +1\}$, and the substitution is linear with the inverse $\nu = d + \sigma u/a$. Explicitly,

$$\begin{cases} d = 0\colon & u = -a\nu \in [0, a] \text{ on } I_-, & u = a\nu \in [0, a] \text{ on } I_+, \\ d = 1\colon & u = a(1 - \nu) \in [a, 2a] \text{ on } I_-, & u = a(1 - \nu) \in [0, a] \text{ on } I_+. \end{cases} \tag{53}$$

In all four cases the singular value $\nu = d$ corresponds to $u = 0$. Every weight $W_{\alpha\beta}$ of (23), restricted to I_s and written in u, is a polynomial in u of degree at most three; these polynomials are listed in Table 2. The y-direction is the same with η, b, d_y and $v = b\,|\eta - d_y|$.

Proof. For $d = 0$ the difference $\nu - d = \nu$ lies in $[-1, 0]$ on I_- and in $[0, 1]$ on I_+, so $\sigma = -1$ and $\sigma = +1$, respectively, and u runs over $[0, a]$ on both. For $d = 1$ the difference $\nu - 1$ lies in $[-2, -1]$ on I_- and in $[-1, 0]$ on I_+, so $\sigma = -1$ on both, and $u = a(1 - \nu)$ runs over $[a, 2a]$ and $[0, a]$. This gives (53). On I_s every weight is one of the polynomials of (23), of degree at most three in ν, and the linear substitution $\nu = d + \sigma u/a$ keeps it a polynomial of the same degree in u. Expanding gives Table 2.

□

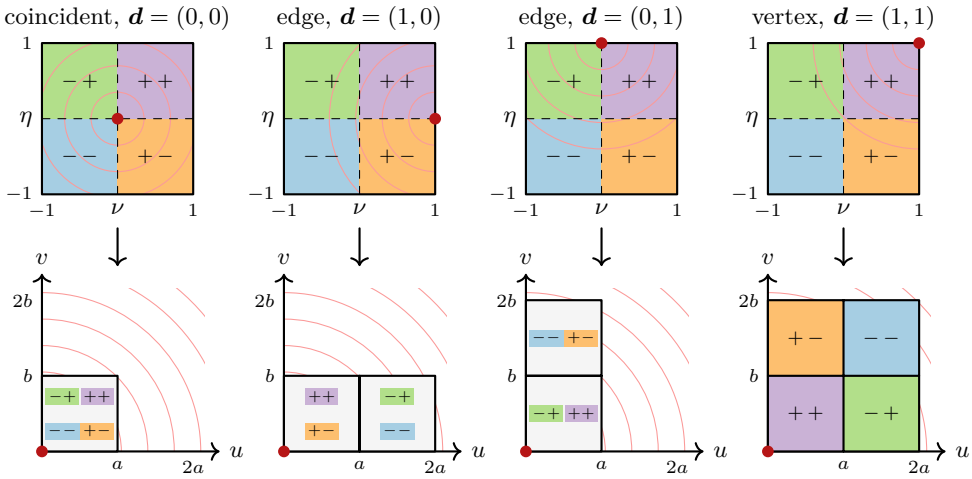


Figure 6: The integration regions of the four touching configurations ($a = b$). Top: the domain $[-1,1]^2$ of Theorem (4.1), split into the pieces $I_s \times I_t$, labelled $s\,t$; the singular point $(\nu,\eta) = (d_x, d_y)$ (red) is the centre, the midpoint of an edge or a corner, and the circles are level lines of $\rho_{\boldsymbol{d}}$. Bottom: the same pieces in the variables $u = \sigma_s a(\nu - d_x)$, $v = \sigma_t b(\eta - d_y)$ of Lemma (6.1). In a direction with $d = 0$ both pieces fold onto $[0,a]$; with $d = 1$ they lie side by side on $[0,a]$ and $[a,2a]$. The singular point is the origin, a corner of every piece that contains it.

Theorem 6.1 (Touching moments). Let $p \in \{-1,1,3\}$, $\boldsymbol{d} = (d_x, d_y) \in \{0,1\}^2$ and $\alpha,\beta \in \{0,1\}$. On the pieces $s,t \in \{-,+\}$ write, with Table 2,

$$W_{\alpha\beta}\big|_{I_s} = \sum_{m=0}^{3} c_m^s\, u^m, \qquad W_{00}\big|_{I_t} = \sum_{n=0}^{1} e_n^t\, v^n, \tag{54}$$

and let $A_s \times B_t$ be the image of the piece, with A_s equal to $[0,a]$ or $[a,2a]$ according to d_x and B_t equal to $[0,b]$ or $[b,2b]$ according to d_y. Then

$$M_{\alpha\beta}^{(p)}(\boldsymbol{d}) = \frac{1}{ab} \sum_{s,t\in\{-,+\}} \sum_{m=0}^{3} \sum_{n=0}^{1} c_m^s\, e_n^t\, \Delta_{A_s\times B_t} G_{mn}^{(p)}, \tag{55}$$

where, with $G = G_{mn}^{(p)}$, which vanishes on the axes,

$$\begin{aligned} \Delta_{[0,a]\times[0,b]} G &= G(a,b), \\ \Delta_{[a,2a]\times[0,b]} G &= G(2a,b) - G(a,b), \\ \Delta_{[0,a]\times[b,2b]} G &= G(a,2b) - G(a,b), \\ \Delta_{[a,2a]\times[b,2b]} G &= G(2a,2b) - G(2a,b) - G(a,2b) + G(a,b). \end{aligned} \tag{56}$$

Every touching moment is therefore an exact, finite combination of $G_{mn}^{(p)}$ at the four points (a,b), $(2a,b)$, $(a,2b)$, $(2a,2b)$. The N-moments follow from Lemma (2.1)(iii) with a and b exchanged.

Proof. By Theorem (4.1), $M_{\alpha\beta}^{(p)}(\boldsymbol{d})$ is the sum over the four pieces $I_s \times I_t$. On a piece substitute $u = \sigma_s a(\nu - d_x)$ and $v = \sigma_t b(\eta - d_y)$: then $\mathrm{d}\nu\,\mathrm{d}\eta = \mathrm{d}u\,\mathrm{d}v/(ab)$, $\rho_{\boldsymbol{d}} = \sqrt{u^2+v^2}$, the weights become the polynomials of Lemma (6.1), and the piece is

Table 2: The weights (23) on the pieces of Lemma (6.1) (x-direction).

d_x	0	0	1	1
piece	I_-, $\sigma=-1$	I_+, $\sigma=+1$	I_-, $\sigma=-1$	I_+, $\sigma=-1$
u	$[0,\,a]$	$[0,\,a]$	$[a,\,2a]$	$[0,\,a]$
W_{00}	$\dfrac{a-u}{a}$	$\dfrac{a-u}{a}$	$\dfrac{2a-u}{a}$	$\dfrac{u}{a}$
W_{10}	$\dfrac{(a-u)^2}{2a^2}$	$\dfrac{(a-u)\,(a+u)}{2a^2}$	$\dfrac{(2a-u)^2}{2a^2}$	$\dfrac{u\,(2a-u)}{2a^2}$
W_{01}	$\dfrac{(a-u)\,(a+u)}{2a^2}$	$\dfrac{(a-u)^2}{2a^2}$	$\dfrac{u\,(2a-u)}{2a^2}$	$\dfrac{u^2}{2a^2}$
W_{11}	$\dfrac{(2a+u)\,(a-u)^2}{6a^3}$	$\dfrac{(2a+u)\,(a-u)^2}{6a^3}$	$\dfrac{(a+u)\,(2a-u)^2}{6a^3}$	$\dfrac{u^2\,(3a-u)}{6a^3}$

$\int_{A_s}\int_{B_t}\sum_{m,n}c_m^s e_n^t u^m v^n (u^2+v^2)^{p/2}\,\mathrm{d}v\,\mathrm{d}u$. Apply (40) term by term; the corners on the axes contribute zero by Proposition (5.1)(i). Each piece is absolutely integrable ($p>-2$), so no limit is involved.

□

Remark. The same corner-sum construction works for every offset, but for distant cells it gives a sum of large, nearly cancelling terms, whereas GL converges exponentially there. It is meant for the touching offsets only.

Corollary 6.1 (Homogeneity). $M^{(p)}_{\alpha\beta}(\boldsymbol{d};\lambda a,\lambda b)=\lambda^p\,M^{(p)}_{\alpha\beta}(\boldsymbol{d};a,b)$ for $\lambda>0$, so the static moments of a grid depend, up to the factor a^p, on the aspect ratio b/a only.

Proof. In (28), $\rho_{\boldsymbol{d}}$ scales by λ and the weights do not depend on a and b.

□

Proposition 6.1 (Agreement with the coincident-cell formulas of the thesis). Consider the integrals $\mathcal{I}^{(-1)}$ and $\mathcal{I}^{(1)}$ of the appendix of [1]. Then

$$\begin{aligned}
(ab)^2 M^{(-1)}_{00}(\mathbf{0}) &= \mathcal{I}^{(-1)}[1],\\
(ab)^2 M^{(-1)}_{11}(\mathbf{0}) &= \mathcal{I}^{(-1)}[\lambda_1\lambda_1'],\\
(ab)^2\big(M^{(-1)}_{10}(\mathbf{0})-M^{(-1)}_{11}(\mathbf{0})\big) &= \mathcal{I}^{(-1)}[\lambda_1(1-\lambda_1')],\\
(ab)^2 M^{(1)}_{00}(\mathbf{0}) &= \mathcal{I}^{(1)}[1],\\
(ab)^2 M^{(1)}_{11}(\mathbf{0}) &= \mathcal{I}^{(1)}[\lambda_1\lambda_1'].
\end{aligned} \tag{57}$$

The first one reads

$$\begin{aligned}
M^{(-1)}_{00}(\mathbf{0}) = \frac{1}{3a^2b^2}\Bigg(&2a^3+2b^3-2\sqrt{a^2+b^2}\,\big(a^2+b^2\big)\\
&+6a^2b\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right)+6ab^2\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\Bigg).
\end{aligned} \tag{58}$$

Proof. By Corollary (6.1) it suffices to take $a = 1$ and keep $b > 0$ symbolic. The difference of the two sides, written with (55) and the cited formula, simplifies to zero exactly (SymPy).

□

Remark (thesis appendix). The values of the formulas in the appendix of [1] are correct, but three labels are misprinted: the integrands of the $\mathcal{I}^{(1)}$ entries are written with $1/R$ instead of R; the last $\mathcal{I}^{(1)}$ entry repeats the label $[\lambda_1(1-\lambda_1')]$ instead of $[\lambda_2(1-\lambda_2')]$; and the integrand of $\mathcal{I}^{(-1)}[\lambda_2(1-\lambda_2')]$ contains $(u+\lambda_2')$ instead of $(v+\lambda_2')$.

A The mixed antiderivatives $G_{mn}^{(p)}$

This appendix lists the closed forms (50) of the antiderivatives (38) for $p \in \{-1, 1, 3\}$, $m \le 3$ and $n \le 1$, which are the ones Theorem (6.1) needs. Any other one follows from $G_{nm}^{(p)}(b,a) = G_{mn}^{(p)}(a,b)$, and on the axes $G_{mn}^{(p)} = 0$.

Kernel R^{-1}.

$$G_{00}^{(-1)}(a,b) = a\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) + b\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)$$

$$G_{01}^{(-1)}(a,b) = \frac{1}{2}\left(-a^2 + a\sqrt{a^2+b^2} + b^2\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\right)$$

$$G_{10}^{(-1)}(a,b) = \frac{1}{2}\left(-b^2 + b\sqrt{a^2+b^2} + a^2\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right)\right)$$

$$G_{11}^{(-1)}(a,b) = \frac{1}{3}\left(-a^3 - b^3 + \sqrt{a^2+b^2}\,(a^2+b^2)\right)$$

$$G_{20}^{(-1)}(a,b) = \frac{1}{6}\left(ab\sqrt{a^2+b^2} + 2a^3\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) - b^3\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\right)$$

$$G_{21}^{(-1)}(a,b) = \frac{1}{8}\left(-2a^4 + a\sqrt{a^2+b^2}\,(2a^2+b^2) - b^4\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\right)$$

$$G_{30}^{(-1)}(a,b) = \frac{1}{12}\left(2b^4 + b\sqrt{a^2+b^2}\,(a^2-2b^2) + 3a^4\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right)\right)$$

$$G_{31}^{(-1)}(a,b) = \frac{1}{15}\left(-3a^5 + 2b^5 + \sqrt{a^2+b^2}\,(3a^4 + a^2b^2 - 2b^4)\right)$$

Kernel R^{1}.

$$G_{00}^{(1)}(a,b) = \frac{1}{6}\left(2ab\sqrt{a^2+b^2} + a^3\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right)\right.$$

$$+ b^3 \log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\Bigg)$$

$$G_{01}^{(1)}(a,b) = \frac{1}{24}\Bigg(-2a^4 + a\sqrt{a^2+b^2}\left(2a^2+5b^2\right) + 3b^4 \log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\Bigg)$$

$$G_{10}^{(1)}(a,b) = \frac{1}{24}\Bigg(-2b^4 + b\sqrt{a^2+b^2}\left(5a^2+2b^2\right) + 3a^4 \log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right)\Bigg)$$

$$G_{11}^{(1)}(a,b) = \frac{1}{15}\left(-a^5 - b^5 + \sqrt{a^2+b^2}\left(a^4+2a^2b^2+b^4\right)\right)$$

$$G_{20}^{(1)}(a,b) = \frac{1}{40}\Bigg(ab\sqrt{a^2+b^2}\left(6a^2+b^2\right) + 4a^5 \log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) - b^5 \log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\Bigg)$$

$$G_{21}^{(1)}(a,b) = \frac{1}{144}\Bigg(-8a^6 + a\sqrt{a^2+b^2}\left(8a^4+14a^2b^2+3b^4\right) - 3b^6 \log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\Bigg)$$

$$G_{30}^{(1)}(a,b) = \frac{1}{180}\Bigg(4b^6 + b\sqrt{a^2+b^2}\left(21a^4+2a^2b^2-4b^4\right) + 15a^6 \log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right)\Bigg)$$

$$G_{31}^{(1)}(a,b) = \frac{1}{105}\left(-5a^7 + 2b^7 + \sqrt{a^2+b^2}\left(5a^6+8a^4b^2+a^2b^4-2b^6\right)\right)$$

Kernel R^3.

$$G_{00}^{(3)}(a,b) = \frac{1}{40}\Bigg(7ab\sqrt{a^2+b^2}\left(a^2+b^2\right) + 3a^5 \log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) + 3b^5 \log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\Bigg)$$

$$G_{01}^{(3)}(a,b) = \frac{1}{240}\Bigg(-8a^6 + a\sqrt{a^2+b^2}\left(8a^4+26a^2b^2+33b^4\right)$$

$$+ 15b^6 \log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\Bigg)$$

$$G_{10}^{(3)}(a,b) = \frac{1}{240}\Bigg(-8b^6 + b\sqrt{a^2+b^2}\left(33a^4+26a^2b^2+8b^4\right)$$
$$+ 15a^6 \log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right)\Bigg)$$

$$G_{11}^{(3)}(a,b) = \frac{1}{35}\Bigg(-a^7-b^7+\sqrt{a^2+b^2}\left(a^6+3a^4b^2+3a^2b^4+b^6\right)\Bigg)$$

$$G_{20}^{(3)}(a,b) = \frac{1}{336}\Bigg(ab\sqrt{a^2+b^2}\left(38a^4+26a^2b^2+3b^4\right)$$
$$+ 18a^7 \log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) - 3b^7 \log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\Bigg)$$

$$G_{21}^{(3)}(a,b) = \frac{1}{1920}\Bigg(-48a^8$$
$$+ a\sqrt{a^2+b^2}\left(48a^6+136a^4b^2+118a^2b^4+15b^6\right)$$
$$- 15b^8 \log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\Bigg)$$

$$G_{30}^{(3)}(a,b) = \frac{1}{2240}\Bigg(16b^8$$
$$+ b\sqrt{a^2+b^2}\left(215a^6+134a^4b^2+8a^2b^4-16b^6\right)$$
$$+ 105a^8 \log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right)\Bigg)$$

$$G_{31}^{(3)}(a,b) = \frac{1}{315}\Bigg(-7a^9+2b^9$$
$$+ \sqrt{a^2+b^2}\left(7a^8+19a^6b^2+15a^4b^4+a^2b^6-2b^8\right)\Bigg)$$

B The touching moments

This appendix lists the x-moments $M_{\alpha\beta}^{(p)}(\boldsymbol{d})$ of the cell $a \times b$ for $p \in \{-1,1,3\}$, $\alpha,\beta \in \{0,1\}$ and the touching offsets $\boldsymbol{d} \in \{(0,0),(1,0),(0,1),(1,1)\}$. They are the corner sums (55) expanded with SymPy, and each one was checked against its corner sum to 30 digits. The N-moments are $N_{\alpha\beta}^{(p)}(d_x,d_y;a,b) = M_{\alpha\beta}^{(p)}(d_y,d_x;b,a)$, and the negative offsets follow from Lemma (2.1)(i) and (ii). For evaluation in double precision

the corner sum (55) is the better form; the expanded formulas are meant for reading and checking.

Kernel R^{-1}

Coincident cell, $\boldsymbol{d} = (0,0)$.

$$M_{00}^{(-1)}(0,0) = \frac{1}{3a^2b^2}\Bigg(2a^3+2b^3-2\sqrt{a^2+b^2}\,(a^2+b^2) + 6a^2b\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) + 6ab^2\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\Bigg)$$

$$M_{10}^{(-1)}(0,0) = \frac{1}{3a^2b^2}\Bigg(a^3+b^3-\sqrt{a^2+b^2}\,(a^2+b^2) + 3a^2b\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) + 3ab^2\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\Bigg)$$

$$M_{01}^{(-1)}(0,0) = \frac{1}{3a^2b^2}\Bigg(a^3+b^3-\sqrt{a^2+b^2}\,(a^2+b^2) + 3a^2b\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) + 3ab^2\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\Bigg)$$

$$M_{11}^{(-1)}(0,0) = \frac{1}{90a^4b^2}\Bigg(12a^5+30a^2b^3+2b^5 - \sqrt{a^2+b^2}\,(12a^4+29a^2b^2+2b^4) + 45a^4b\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) + 60a^3b^2\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\Bigg)$$

Shared edge parallel to y, $\boldsymbol{d} = (1,0)$.

$$M_{00}^{(-1)}(1,0) = \frac{1}{3a^2b^2}\Bigg(6a^3-b^3+2\sqrt{a^2+b^2}\,(a^2+b^2) - \sqrt{4a^2+b^2}\,(4a^2+b^2) + 12a^2b\log\left(\frac{\sqrt{4a^2+b^2}+b}{2a}\right) - 6a^2b\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) - 6ab^2\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right) + 6ab^2\log\left(\frac{\sqrt{4a^2+b^2}+2a}{b}\right)\Bigg)$$

$$M_{10}^{(-1)}(1,0) = \frac{1}{24a^3b^2}\Bigg(20a^4-8ab^3+2a\sqrt{a^2+b^2}\,(6a^2+11b^2)$$

$$
\begin{aligned}
&- 2a\sqrt{4a^2+b^2}\left(8a^2+7b^2\right)\\
&+ 64a^3b\log\left(\frac{\sqrt{4a^2+b^2}+b}{2a}\right)\\
&- 40a^3b\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right)\\
&- 2b^2\left(24a^2-b^2\right)\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\\
&+ b^2\left(48a^2-b^2\right)\log\left(\frac{\sqrt{4a^2+b^2}+2a}{b}\right)\Bigg)
\end{aligned}
$$

$$
\begin{aligned}
M_{01}^{(-1)}(1,0) = \frac{1}{24a^3b^2}\Bigg(&28a^4+2a\sqrt{a^2+b^2}\left(2a^2-3b^2\right)\\
&- 2a\sqrt{4a^2+b^2}\left(8a^2-3b^2\right)\\
&+ 32a^3b\log\left(\frac{\sqrt{4a^2+b^2}+b}{2a}\right) - 8a^3b\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right)\\
&- 2b^4\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right) + b^4\log\left(\frac{\sqrt{4a^2+b^2}+2a}{b}\right)\Bigg)
\end{aligned}
$$

$$
\begin{aligned}
M_{11}^{(-1)}(1,0) = \frac{1}{360a^4b^2}\Bigg(&180a^5-4b^5+2\sqrt{a^2+b^2}\left(54a^4+13a^2b^2+4b^4\right)\\
&- 2\sqrt{4a^2+b^2}\left(72a^4+11a^2b^2+2b^4\right)\\
&+ 480a^4b\log\left(\frac{\sqrt{4a^2+b^2}+b}{2a}\right)\\
&- 300a^4b\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right)\\
&- 30ab^2\left(8a^2+b^2\right)\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\\
&+ 15ab^2\left(16a^2+b^2\right)\log\left(\frac{\sqrt{4a^2+b^2}+2a}{b}\right)\Bigg)
\end{aligned}
$$

Shared edge parallel to x, $\boldsymbol{d} = (0,1)$.

$$
\begin{aligned}
M_{00}^{(-1)}(0,1) = \frac{1}{3a^2b^2}\Bigg(&-a^3+6b^3+2\sqrt{a^2+b^2}\left(a^2+b^2\right)\\
&- \sqrt{a^2+4b^2}\left(a^2+4b^2\right) - 6a^2b\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right)
\end{aligned}
$$

$$
\begin{aligned}
&+ 6a^2 b \log\left(\frac{\sqrt{a^2+4b^2}+2b}{a}\right) \\
&+ 12ab^2 \log\left(\frac{\sqrt{a^2+4b^2}+a}{2b}\right) - 6ab^2 \log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\Bigg)
\end{aligned}
$$

$$
\begin{aligned}
M_{10}^{(-1)}(0,1) = \frac{1}{6a^2b^2}\Bigg(&- a^3 + 6b^3 + 2\sqrt{a^2+b^2}\left(a^2+b^2\right) \\
&- \sqrt{a^2+4b^2}\left(a^2+4b^2\right) - 6a^2 b \log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) \\
&+ 6a^2 b \log\left(\frac{\sqrt{a^2+4b^2}+2b}{a}\right) \\
&+ 12ab^2 \log\left(\frac{\sqrt{a^2+4b^2}+a}{2b}\right) - 6ab^2 \log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\Bigg)
\end{aligned}
$$

$$
\begin{aligned}
M_{01}^{(-1)}(0,1) = \frac{1}{6a^2b^2}\Bigg(&- a^3 + 6b^3 + 2\sqrt{a^2+b^2}\left(a^2+b^2\right) \\
&- \sqrt{a^2+4b^2}\left(a^2+4b^2\right) - 6a^2 b \log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) \\
&+ 6a^2 b \log\left(\frac{\sqrt{a^2+4b^2}+2b}{a}\right) \\
&+ 12ab^2 \log\left(\frac{\sqrt{a^2+4b^2}+a}{2b}\right) - 6ab^2 \log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\Bigg)
\end{aligned}
$$

$$
\begin{aligned}
M_{11}^{(-1)}(0,1) = \frac{1}{90a^4b^2}\Bigg(&- 6a^5 + 90a^2b^3 + 30b^5 \\
&+ \sqrt{a^2+b^2}\left(12a^4 + 29a^2b^2 + 2b^4\right) \\
&- 2\sqrt{a^2+4b^2}\left(3a^4 + 29a^2b^2 + 8b^4\right) \\
&- 45a^4 b \log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) \\
&+ 45a^4 b \log\left(\frac{\sqrt{a^2+4b^2}+2b}{a}\right) \\
&+ 120a^3b^2 \log\left(\frac{\sqrt{a^2+4b^2}+a}{2b}\right) \\
&- 60a^3b^2 \log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\Bigg)
\end{aligned}
$$

Shared vertex, $\boldsymbol{d} = (1,1)$.

$$
\begin{aligned}
M_{00}^{(-1)}(1,1) = \frac{1}{3a^2b^2}\Bigg(& -3a^3 - 3b^3 - 6\sqrt{a^2+b^2}\left(a^2+b^2\right) \\
& + \sqrt{a^2+4b^2}\left(a^2+4b^2\right) + \sqrt{4a^2+b^2}\left(4a^2+b^2\right) \\
& - 12a^2b\log\left(\frac{\sqrt{4a^2+b^2}+b}{2a}\right) \\
& + 18a^2b\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) \\
& - 6a^2b\log\left(\frac{\sqrt{a^2+4b^2}+2b}{a}\right) \\
& - 12ab^2\log\left(\frac{\sqrt{a^2+4b^2}+a}{2b}\right) \\
& + 18ab^2\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right) \\
& - 6ab^2\log\left(\frac{\sqrt{4a^2+b^2}+2a}{b}\right)\Bigg)
\end{aligned}
$$

$$
\begin{aligned}
M_{10}^{(-1)}(1,1) = \frac{1}{24a^3b^2}\Bigg(& -10a^4 - 24ab^3 - 2a\sqrt{a^2+b^2}\left(14a^2+39b^2\right) \\
& + 2a\sqrt{a^2+4b^2}\left(3a^2+22b^2\right) \\
& + 2a\sqrt{4a^2+b^2}\left(8a^2+7b^2\right) \\
& - 64a^3b\log\left(\frac{\sqrt{4a^2+b^2}+b}{2a}\right) \\
& + 104a^3b\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) \\
& - 40a^3b\log\left(\frac{\sqrt{a^2+4b^2}+2b}{a}\right) \\
& - 16b^2\left(6a^2-b^2\right)\log\left(\frac{\sqrt{a^2+4b^2}+a}{2b}\right) \\
& + 2b^2\left(72a^2-5b^2\right)\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right) \\
& - b^2\left(48a^2-b^2\right)\log\left(\frac{\sqrt{4a^2+b^2}+2a}{b}\right)\Bigg)
\end{aligned}
$$

$$
\begin{aligned}
M_{01}^{(-1)}(1,1) = \frac{1}{24a^3b^2}\Bigg(& -14a^4 - 10a\sqrt{a^2+b^2}\left(2a^2-3b^2\right) \\
& + 2a\sqrt{a^2+4b^2}\left(a^2-6b^2\right) \\
& + 2a\sqrt{4a^2+b^2}\left(8a^2-3b^2\right) \\
& - 32a^3b\log\left(\frac{\sqrt{4a^2+b^2}+b}{2a}\right) \\
& + 40a^3b\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) \\
& - 8a^3b\log\left(\frac{\sqrt{a^2+4b^2}+2b}{a}\right) \\
& - 16b^4\log\left(\frac{\sqrt{a^2+4b^2}+a}{2b}\right) + 10b^4\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right) \\
& - b^4\log\left(\frac{\sqrt{4a^2+b^2}+2a}{b}\right)\Bigg)
\end{aligned}
$$

$$
\begin{aligned}
M_{11}^{(-1)}(1,1) = \frac{1}{360a^4b^2}\Bigg(& -90a^5 - 60b^5 - 6\sqrt{a^2+b^2}\left(42a^4+19a^2b^2+12b^4\right) \\
& + 2\sqrt{a^2+4b^2}\left(27a^4+26a^2b^2+32b^4\right) \\
& + 2\sqrt{4a^2+b^2}\left(72a^4+11a^2b^2+2b^4\right) \\
& - 480a^4b\log\left(\frac{\sqrt{4a^2+b^2}+b}{2a}\right) \\
& + 780a^4b\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) \\
& - 300a^4b\log\left(\frac{\sqrt{a^2+4b^2}+2b}{a}\right) \\
& - 240ab^2\left(2a^2+b^2\right)\log\left(\frac{\sqrt{a^2+4b^2}+a}{2b}\right) \\
& + 30ab^2\left(24a^2+5b^2\right)\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right) \\
& - 15ab^2\left(16a^2+b^2\right)\log\left(\frac{\sqrt{4a^2+b^2}+2a}{b}\right)\Bigg)
\end{aligned}
$$

Kernel R^1

Coincident cell, $\boldsymbol{d} = (0,0)$.

$$M_{00}^{(1)}(0,0) = \frac{1}{30a^2b^2}\Bigg(2a^5 + 2b^5 - 2\sqrt{a^2+b^2}\left(a^4 - 3a^2b^2 + b^4\right) + 5a^4b\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) + 5ab^4\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\Bigg)$$

$$M_{10}^{(1)}(0,0) = \frac{1}{60a^2b^2}\Bigg(2a^5 + 2b^5 - 2\sqrt{a^2+b^2}\left(a^4 - 3a^2b^2 + b^4\right) + 5a^4b\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) + 5ab^4\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\Bigg)$$

$$M_{01}^{(1)}(0,0) = \frac{1}{60a^2b^2}\Bigg(2a^5 + 2b^5 - 2\sqrt{a^2+b^2}\left(a^4 - 3a^2b^2 + b^4\right) + 5a^4b\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) + 5ab^4\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\Bigg)$$

$$M_{11}^{(1)}(0,0) = \frac{1}{3780a^4b^2}\Bigg(36a^7 + 126a^2b^5 + 8b^7 - \sqrt{a^2+b^2}\left(36a^6 - 165a^4b^2 + 122a^2b^4 + 8b^6\right) + 105a^6b\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) + 210a^3b^4\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\Bigg)$$

Shared edge parallel to y, $\boldsymbol{d} = (1,0)$.

$$M_{00}^{(1)}(1,0) = \frac{1}{30a^2b^2}\Bigg(30a^5 - b^5 + 2\sqrt{a^2+b^2}\left(a^4 - 3a^2b^2 + b^4\right) - \sqrt{4a^2+b^2}\left(16a^4 - 12a^2b^2 + b^4\right) + 40a^4b\log\left(\frac{\sqrt{4a^2+b^2}+b}{2a}\right) - 5a^4b\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) - 5ab^4\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right) + 5ab^4\log\left(\frac{\sqrt{4a^2+b^2}+2a}{b}\right)\Bigg)$$

$$M_{10}^{(1)}(1,0) = \frac{1}{720a^3b^2}\Bigg(224a^6 - 24ab^5 + 2a\sqrt{a^2+b^2}\left(16a^4 - 56a^2b^2 + 33b^4\right)$$

$$
\begin{aligned}
&- 2a\sqrt{4a^2+b^2}\left(64a^4-80a^2b^2+21b^4\right)\\
&+ 384a^5b\log\left(\frac{\sqrt{4a^2+b^2}+b}{2a}\right)\\
&- 84a^5b\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right)\\
&- 6b^4\left(20a^2-b^2\right)\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\\
&+ 3b^4\left(40a^2-b^2\right)\log\left(\frac{\sqrt{4a^2+b^2}+2a}{b}\right)\Bigg)
\end{aligned}
$$

$$
\begin{aligned}
M_{01}^{(1)}(1,0) = \frac{1}{720a^3b^2}\Bigg(&496a^6+2a\sqrt{a^2+b^2}\left(8a^4-16a^2b^2-9b^4\right)\\
&- 2a\sqrt{4a^2+b^2}\left(128a^4-64a^2b^2-9b^4\right)\\
&+ 576a^5b\log\left(\frac{\sqrt{4a^2+b^2}+b}{2a}\right)\\
&- 36a^5b\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) - 6b^6\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\\
&+ 3b^6\log\left(\frac{\sqrt{4a^2+b^2}+2a}{b}\right)\Bigg)
\end{aligned}
$$

$$
\begin{aligned}
M_{11}^{(1)}(1,0) = \frac{1}{15120a^4b^2}\Bigg(&3360a^7-16b^7\\
&+ 2\sqrt{a^2+b^2}\left(240a^6-666a^4b^2+55a^2b^4+16b^6\right)\\
&- 2\sqrt{4a^2+b^2}\left(960a^6-960a^4b^2+47a^2b^4+8b^6\right)\\
&+ 5376a^6b\log\left(\frac{\sqrt{4a^2+b^2}+b}{2a}\right)\\
&- 1176a^6b\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right)\\
&- 42ab^4\left(20a^2+3b^2\right)\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\\
&+ 21ab^4\left(40a^2+3b^2\right)\log\left(\frac{\sqrt{4a^2+b^2}+2a}{b}\right)\Bigg)
\end{aligned}
$$

Shared edge parallel to x, $\boldsymbol{d} = (0,1)$.

$$
\begin{aligned}
M_{00}^{(1)}(0,1) = \frac{1}{30a^2b^2}\Bigg(& -a^5 + 30b^5 + 2\sqrt{a^2+b^2}\left(a^4 - 3a^2b^2 + b^4\right) \\
& - \sqrt{a^2+4b^2}\left(a^4 - 12a^2b^2 + 16b^4\right) \\
& - 5a^4 b \log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) + 5a^4 b \log\left(\frac{\sqrt{a^2+4b^2}+2b}{a}\right) \\
& + 40ab^4 \log\left(\frac{\sqrt{a^2+4b^2}+a}{2b}\right) - 5ab^4 \log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\Bigg)
\end{aligned}
$$

$$
\begin{aligned}
M_{10}^{(1)}(0,1) = \frac{1}{60a^2b^2}\Bigg(& -a^5 + 30b^5 + 2\sqrt{a^2+b^2}\left(a^4 - 3a^2b^2 + b^4\right) \\
& - \sqrt{a^2+4b^2}\left(a^4 - 12a^2b^2 + 16b^4\right) \\
& - 5a^4 b \log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) + 5a^4 b \log\left(\frac{\sqrt{a^2+4b^2}+2b}{a}\right) \\
& + 40ab^4 \log\left(\frac{\sqrt{a^2+4b^2}+a}{2b}\right) - 5ab^4 \log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\Bigg)
\end{aligned}
$$

$$
\begin{aligned}
M_{01}^{(1)}(0,1) = \frac{1}{60a^2b^2}\Bigg(& -a^5 + 30b^5 + 2\sqrt{a^2+b^2}\left(a^4 - 3a^2b^2 + b^4\right) \\
& - \sqrt{a^2+4b^2}\left(a^4 - 12a^2b^2 + 16b^4\right) \\
& - 5a^4 b \log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) + 5a^4 b \log\left(\frac{\sqrt{a^2+4b^2}+2b}{a}\right) \\
& + 40ab^4 \log\left(\frac{\sqrt{a^2+4b^2}+a}{2b}\right) - 5ab^4 \log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\Bigg)
\end{aligned}
$$

$$
\begin{aligned}
M_{11}^{(1)}(0,1) = \frac{1}{3780a^4b^2}\Bigg(& -18a^7 + 1890a^2b^5 + 504b^7 \\
& + \sqrt{a^2+b^2}\left(36a^6 - 165a^4b^2 + 122a^2b^4 + 8b^6\right) \\
& - 2\sqrt{a^2+4b^2}\left(9a^6 - 165a^4b^2 + 488a^2b^4 + 128b^6\right) \\
& - 105a^6 b \log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) \\
& + 105a^6 b \log\left(\frac{\sqrt{a^2+4b^2}+2b}{a}\right)
\end{aligned}
$$

$$+ 1680a^3b^4 \log\left(\frac{\sqrt{a^2+4b^2}+a}{2b}\right)$$
$$- 210a^3b^4 \log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\Bigg)$$

Shared vertex, $\boldsymbol{d} = (1,1)$.

$$M_{00}^{(1)}(1,1) = \frac{1}{30a^2b^2}\Bigg(- 15a^5 - 15b^5 - 18\sqrt{a^2+b^2}\left(a^4 - 3a^2b^2 + b^4\right)$$
$$+ \sqrt{a^2+4b^2}\left(a^4 - 12a^2b^2 + 16b^4\right)$$
$$+ \sqrt{4a^2+b^2}\left(16a^4 - 12a^2b^2 + b^4\right)$$
$$- 40a^4b \log\left(\frac{\sqrt{4a^2+b^2}+b}{2a}\right)$$
$$+ 45a^4b \log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right)$$
$$- 5a^4b \log\left(\frac{\sqrt{a^2+4b^2}+2b}{a}\right)$$
$$- 40ab^4 \log\left(\frac{\sqrt{a^2+4b^2}+a}{2b}\right)$$
$$+ 45ab^4 \log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)$$
$$- 5ab^4 \log\left(\frac{\sqrt{4a^2+b^2}+2a}{b}\right)\Bigg)$$

$$M_{10}^{(1)}(1,1) = \frac{1}{720a^3b^2}\Bigg(- 112a^6 - 360ab^5$$
$$- 2a\sqrt{a^2+b^2}\left(80a^4 - 376a^2b^2 + 369b^4\right)$$
$$+ 16a\sqrt{a^2+4b^2}\left(a^4 - 14a^2b^2 + 33b^4\right)$$
$$+ 2a\sqrt{4a^2+b^2}\left(64a^4 - 80a^2b^2 + 21b^4\right)$$
$$- 384a^5b \log\left(\frac{\sqrt{4a^2+b^2}+b}{2a}\right)$$
$$+ 468a^5b \log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right)$$

$$
\begin{aligned}
&- 84a^5 b \log\left(\frac{\sqrt{a^2+4b^2}+2b}{a}\right)\\
&- 192b^4\left(5a^2-b^2\right)\log\left(\frac{\sqrt{a^2+4b^2}+a}{2b}\right)\\
&+ 6b^4\left(180a^2-17b^2\right)\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\\
&- 3b^4\left(40a^2-b^2\right)\log\left(\frac{\sqrt{4a^2+b^2}+2a}{b}\right)\Bigg)
\end{aligned}
$$

$$
\begin{aligned}
M_{01}^{(1)}(1,1) = \frac{1}{720a^3b^2}\Bigg(&- 248a^6 - 34a\sqrt{a^2+b^2}\left(8a^4-16a^2b^2-9b^4\right)\\
&+ 8a\sqrt{a^2+4b^2}\left(a^4-8a^2b^2-18b^4\right)\\
&+ 2a\sqrt{4a^2+b^2}\left(128a^4-64a^2b^2-9b^4\right)\\
&- 576a^5 b\log\left(\frac{\sqrt{4a^2+b^2}+b}{2a}\right)\\
&+ 612a^5 b\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right)\\
&- 36a^5 b\log\left(\frac{\sqrt{a^2+4b^2}+2b}{a}\right)\\
&- 192b^6\log\left(\frac{\sqrt{a^2+4b^2}+a}{2b}\right)\\
&+ 102b^6\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right) - 3b^6\log\left(\frac{\sqrt{4a^2+b^2}+2a}{b}\right)\Bigg)
\end{aligned}
$$

$$
\begin{aligned}
M_{11}^{(1)}(1,1) = \frac{1}{15120a^4b^2}\Bigg(&- 1680a^7 - 1008b^7 - 6\sqrt{a^2+b^2}\big(400a^6-1502a^4b^2\\
&+ 269a^2b^4+176b^6\big)\\
&+ 8\sqrt{a^2+4b^2}\left(30a^6-333a^4b^2+110a^2b^4+128b^6\right)\\
&+ 2\sqrt{4a^2+b^2}\left(960a^6-960a^4b^2+47a^2b^4+8b^6\right)\\
&- 5376a^6 b\log\left(\frac{\sqrt{4a^2+b^2}+b}{2a}\right)\\
&+ 6552a^6 b\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right)
\end{aligned}
$$

$$
\begin{aligned}
&- 1176a^6 b \log\left(\frac{\sqrt{a^2+4b^2}+2b}{a}\right) \\
&- 1344ab^4\left(5a^2+3b^2\right)\log\left(\frac{\sqrt{a^2+4b^2}+a}{2b}\right) \\
&+ 126ab^4\left(60a^2+17b^2\right)\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right) \\
&- 21ab^4\left(40a^2+3b^2\right)\log\left(\frac{\sqrt{4a^2+b^2}+2a}{b}\right)\Bigg)
\end{aligned}
$$

Kernel R^3

Coincident cell, $\boldsymbol{d} = (0,0)$.

$$
\begin{aligned}
M_{00}^{(3)}(0,0) = \frac{1}{420a^2b^2}\Bigg(&8a^7+8b^7 \\
&- \sqrt{a^2+b^2}\left(8a^6-25a^4b^2-25a^2b^4+8b^6\right) \\
&+ 21a^6 b\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right)+21ab^6\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\Bigg)
\end{aligned}
$$

$$
\begin{aligned}
M_{10}^{(3)}(0,0) = \frac{1}{840a^2b^2}\Bigg(&8a^7+8b^7 \\
&- \sqrt{a^2+b^2}\left(8a^6-25a^4b^2-25a^2b^4+8b^6\right) \\
&+ 21a^6 b\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right)+21ab^6\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\Bigg)
\end{aligned}
$$

$$
\begin{aligned}
M_{01}^{(3)}(0,0) = \frac{1}{840a^2b^2}\Bigg(&8a^7+8b^7 \\
&- \sqrt{a^2+b^2}\left(8a^6-25a^4b^2-25a^2b^4+8b^6\right) \\
&+ 21a^6 b\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right)+21ab^6\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\Bigg)
\end{aligned}
$$

$$
\begin{aligned}
M_{11}^{(3)}(0,0) = \frac{1}{30240a^4b^2}\Bigg(&64a^9+288a^2b^7+16b^9-\sqrt{a^2+b^2}\big(64a^8 \\
&- 275a^6b^2-390a^4b^4+280a^2b^6+16b^8\big) \\
&+ 189a^8 b\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right)
\end{aligned}
$$

$$+ 504a^3b^6 \log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\Bigg)$$

Shared edge parallel to y, $\boldsymbol{d} = (1,0)$.

$$\begin{aligned}
M_{00}^{(3)}(1,0) = \frac{1}{420a^2b^2}\Bigg(&504a^7 - 4b^7 \\
&+ \sqrt{a^2+b^2}\left(8a^6 - 25a^4b^2 - 25a^2b^4 + 8b^6\right) \\
&- 2\sqrt{4a^2+b^2}\left(128a^6 - 100a^4b^2 - 25a^2b^4 + 2b^6\right) \\
&+ 672a^6b \log\left(\frac{\sqrt{4a^2+b^2}+b}{2a}\right) \\
&- 21a^6b \log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) - 21ab^6 \log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right) \\
&+ 21ab^6 \log\left(\frac{\sqrt{4a^2+b^2}+2a}{b}\right)\Bigg)
\end{aligned}$$

$$\begin{aligned}
M_{10}^{(3)}(1,0) = \frac{1}{13440a^3b^2}\Bigg(&3936a^8 - 128ab^7 \\
&+ 2a\sqrt{a^2+b^2}\left(80a^6 - 272a^4b^2 - 310a^2b^4 + 177b^6\right) \\
&- 2a\sqrt{4a^2+b^2}\big(1024a^6 - 1152a^4b^2 - 440a^2b^4 \\
&+ 113b^6\big) + 6144a^7b \log\left(\frac{\sqrt{4a^2+b^2}+b}{2a}\right) \\
&- 432a^7b \log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) \\
&- 6b^6\left(112a^2 - 5b^2\right)\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right) \\
&+ 3b^6\left(224a^2 - 5b^2\right)\log\left(\frac{\sqrt{4a^2+b^2}+2a}{b}\right)\Bigg)
\end{aligned}$$

$$\begin{aligned}
M_{01}^{(3)}(1,0) = \frac{1}{13440a^3b^2}\Bigg(&12192a^8 \\
&+ 2a\sqrt{a^2+b^2}\left(48a^6 - 128a^4b^2 - 90a^2b^4 - 49b^6\right) \\
&- 2a\sqrt{4a^2+b^2}\big(3072a^6 - 2048a^4b^2 - 360a^2b^4
\end{aligned}$$

$$
\begin{aligned}
&\quad - 49b^6\big) + 15360a^7 b \log\left(\frac{\sqrt{4a^2+b^2}+b}{2a}\right) \\
&\quad - 240a^7 b \log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) - 30b^8 \log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right) \\
&\quad + 15b^8 \log\left(\frac{\sqrt{4a^2+b^2}+2a}{b}\right)\Bigg)
\end{aligned}
$$

$$
\begin{aligned}
M_{11}^{(3)}(1,0) = \frac{1}{120960a^4b^2}\Bigg(&27552a^9 - 32b^9 + 2\sqrt{a^2+b^2}\big(560a^8 - 1702a^6b^2 \\
&- 1590a^4b^4 + 119a^2b^6 + 32b^8\big) \\
&- 2\sqrt{4a^2+b^2}\big(7168a^8 - 7232a^6b^2 - 2280a^4b^4 \\
&+ 103a^2b^6 + 16b^8\big) + 41472a^8 b \log\left(\frac{\sqrt{4a^2+b^2}+b}{2a}\right) \\
&- 2916a^8 b \log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) \\
&- 18ab^6\left(112a^2+15b^2\right)\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right) \\
&+ 9ab^6\left(224a^2+15b^2\right)\log\left(\frac{\sqrt{4a^2+b^2}+2a}{b}\right)\Bigg)
\end{aligned}
$$

Shared edge parallel to x, $\boldsymbol{d} = (0,1)$.

$$
\begin{aligned}
M_{00}^{(3)}(0,1) = \frac{1}{420a^2b^2}\Bigg(&-4a^7 + 504b^7 \\
&+ \sqrt{a^2+b^2}\left(8a^6 - 25a^4b^2 - 25a^2b^4 + 8b^6\right) \\
&- 2\sqrt{a^2+4b^2}\left(2a^6 - 25a^4b^2 - 100a^2b^4 + 128b^6\right) \\
&- 21a^6 b \log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) \\
&+ 21a^6 b \log\left(\frac{\sqrt{a^2+4b^2}+2b}{a}\right) \\
&+ 672ab^6 \log\left(\frac{\sqrt{a^2+4b^2}+a}{2b}\right) \\
&- 21ab^6 \log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\Bigg)
\end{aligned}
$$

$$M_{10}^{(3)}(0,1) = \frac{1}{840a^2b^2}\Bigg(-4a^7 + 504b^7$$
$$+ \sqrt{a^2+b^2}\left(8a^6 - 25a^4b^2 - 25a^2b^4 + 8b^6\right)$$
$$- 2\sqrt{a^2+4b^2}\left(2a^6 - 25a^4b^2 - 100a^2b^4 + 128b^6\right)$$
$$- 21a^6b\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right)$$
$$+ 21a^6b\log\left(\frac{\sqrt{a^2+4b^2}+2b}{a}\right)$$
$$+ 672ab^6\log\left(\frac{\sqrt{a^2+4b^2}+a}{2b}\right)$$
$$- 21ab^6\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\Bigg)$$

$$M_{01}^{(3)}(0,1) = \frac{1}{840a^2b^2}\Bigg(-4a^7 + 504b^7$$
$$+ \sqrt{a^2+b^2}\left(8a^6 - 25a^4b^2 - 25a^2b^4 + 8b^6\right)$$
$$- 2\sqrt{a^2+4b^2}\left(2a^6 - 25a^4b^2 - 100a^2b^4 + 128b^6\right)$$
$$- 21a^6b\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right)$$
$$+ 21a^6b\log\left(\frac{\sqrt{a^2+4b^2}+2b}{a}\right)$$
$$+ 672ab^6\log\left(\frac{\sqrt{a^2+4b^2}+a}{2b}\right)$$
$$- 21ab^6\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\Bigg)$$

$$M_{11}^{(3)}(0,1) = \frac{1}{30240a^4b^2}\Bigg(-32a^9 + 18144a^2b^7 + 4080b^9 + \sqrt{a^2+b^2}\big(64a^8$$
$$- 275a^6b^2 - 390a^4b^4 + 280a^2b^6 + 16b^8\big)$$
$$- 2\sqrt{a^2+4b^2}\big(16a^8 - 275a^6b^2 - 1560a^4b^4$$
$$+ 4480a^2b^6 + 1024b^8\big) - 189a^8b\log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right)$$

$$
\begin{aligned}
&+ 189a^8 b \log\left(\frac{\sqrt{a^2+4b^2}+2b}{a}\right)\\
&+ 16128a^3b^6 \log\left(\frac{\sqrt{a^2+4b^2}+a}{2b}\right)\\
&- 504a^3b^6 \log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\Bigg)
\end{aligned}
$$

Shared vertex, $\boldsymbol{d} = (1,1)$.

$$
\begin{aligned}
M_{00}^{(3)}(1,1) = \frac{1}{420a^2b^2}\Bigg(&- 252a^7 - 252b^7\\
&- 33\sqrt{a^2+b^2}\left(8a^6 - 25a^4b^2 - 25a^2b^4 + 8b^6\right)\\
&+ 2\sqrt{a^2+4b^2}\left(2a^6 - 25a^4b^2 - 100a^2b^4 + 128b^6\right)\\
&+ 2\sqrt{4a^2+b^2}\left(128a^6 - 100a^4b^2 - 25a^2b^4 + 2b^6\right)\\
&- 672a^6 b \log\left(\frac{\sqrt{4a^2+b^2}+b}{2a}\right)\\
&+ 693a^6 b \log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right)\\
&- 21a^6 b \log\left(\frac{\sqrt{a^2+4b^2}+2b}{a}\right)\\
&- 672ab^6 \log\left(\frac{\sqrt{a^2+4b^2}+a}{2b}\right)\\
&+ 693ab^6 \log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\\
&- 21ab^6 \log\left(\frac{\sqrt{4a^2+b^2}+2a}{b}\right)\Bigg)
\end{aligned}
$$

$$
\begin{aligned}
M_{10}^{(3)}(1,1) = \frac{1}{13440a^3b^2}\Bigg(&- 1968a^8 - 8064ab^7 - 2a\sqrt{a^2+b^2}\big(1104a^6\\
&- 4880a^4b^2 - 7350a^2b^4 + 7409b^6\big)\\
&+ 16a\sqrt{a^2+4b^2}\left(5a^6 - 68a^4b^2 - 310a^2b^4 + 708b^6\right)\\
&+ 2a\sqrt{4a^2+b^2}\big(1024a^6 - 1152a^4b^2 - 440a^2b^4
\end{aligned}
$$

$$
\begin{aligned}
&+ 113b^6\big) - 6144a^7 b \log\left(\frac{\sqrt{4a^2+b^2}+b}{2a}\right)\\
&+ 6576a^7 b \log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right)\\
&- 432a^7 b \log\left(\frac{\sqrt{a^2+4b^2}+2b}{a}\right)\\
&- 768b^6\left(28a^2-5b^2\right)\log\left(\frac{\sqrt{a^2+4b^2}+a}{2b}\right)\\
&+ 6b^6\left(3696a^2-325b^2\right)\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\\
&- 3b^6\left(224a^2-5b^2\right)\log\left(\frac{\sqrt{4a^2+b^2}+2a}{b}\right)\Bigg)
\end{aligned}
$$

$$
\begin{aligned}
M_{01}^{(3)}(1,1) = \frac{1}{13440a^3b^2}\Bigg(&- 6096a^8\\
&- 130a\sqrt{a^2+b^2}\left(48a^6-128a^4b^2-90a^2b^4-49b^6\right)\\
&+ 16a\sqrt{a^2+4b^2}\left(3a^6-32a^4b^2-90a^2b^4-196b^6\right)\\
&+ 2a\sqrt{4a^2+b^2}\big(3072a^6-2048a^4b^2-360a^2b^4\\
&- 49b^6\big) - 15360a^7 b \log\left(\frac{\sqrt{4a^2+b^2}+b}{2a}\right)\\
&+ 15600a^7 b \log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right)\\
&- 240a^7 b \log\left(\frac{\sqrt{a^2+4b^2}+2b}{a}\right)\\
&- 3840b^8 \log\left(\frac{\sqrt{a^2+4b^2}+a}{2b}\right)\\
&+ 1950b^8 \log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right)\\
&- 15b^8 \log\left(\frac{\sqrt{4a^2+b^2}+2a}{b}\right)\Bigg)
\end{aligned}
$$

$$
\begin{aligned}
M_{11}^{(3)}(1,1) = \frac{1}{120960a^4b^2}\Bigg(&- 13776a^9 - 8160b^9 - 6\sqrt{a^2+b^2}\big(2576a^8\\
&- 10210a^6b^2 - 12690a^4b^4 + 2237a^2b^6 + 1376b^8\big)
\end{aligned}
$$

$$
\begin{aligned}
&+ 8\sqrt{a^2+4b^2}\big(70a^8 - 851a^6b^2 - 3180a^4b^4 \\
&+ 952a^2b^6 + 1024b^8\big) + 2\sqrt{4a^2+b^2}\big(7168a^8 \\
&- 7232a^6b^2 - 2280a^4b^4 + 103a^2b^6 + 16b^8\big) \\
&- 41472a^8 b \log\left(\frac{\sqrt{4a^2+b^2}+b}{2a}\right) \\
&+ 44388a^8 b \log\left(\frac{\sqrt{a^2+b^2}+b}{a}\right) \\
&- 2916a^8 b \log\left(\frac{\sqrt{a^2+4b^2}+2b}{a}\right) \\
&- 2304ab^6\left(28a^2+15b^2\right)\log\left(\frac{\sqrt{a^2+4b^2}+a}{2b}\right) \\
&+ 54ab^6\left(1232a^2+325b^2\right)\log\left(\frac{\sqrt{a^2+b^2}+a}{b}\right) \\
&- 9ab^6\left(224a^2+15b^2\right)\log\left(\frac{\sqrt{4a^2+b^2}+2a}{b}\right)\Bigg)
\end{aligned}
$$

References

[1] T. Macháček, "Method of moments for analysis of planar radiators," Bachelor thesis, Czech Technical University in Prague, Faculty of Electrical Engineering, 2025, supervisor M. Čapek.

[2] A. Tzoulis and T. F. Eibert, "Review of singular potential integrals for method of moments solutions of surface integral equations," *Advances in Radio Science*, vol. 2, pp. 93–99, 2004.

[3] S. A. Sauter and C. Schwab, *Boundary Element Methods*, ser. Springer Series in Computational Mathematics. Berlin, Heidelberg: Springer, 2011, vol. 39.

[4] S. Bao, D. Wang, Y. Mo, S. Hu, J. Gu, and W. Tang, "Fully analytical evaluation of singular integrals with RWG and rooftop basis functions," *IEEE Journal on Multiscale and Multiphysics Computational Techniques*, vol. 5, pp. 217–226, 2020.

[5] D. R. Wilton, S. M. Rao, A. W. Glisson, D. H. Schaubert, O. M. Al-Bundak, and C. M. Butler, "Potential integrals for uniform and linear source distributions on polygonal and polyhedral domains," *IEEE Transactions on Antennas and Propagation*, vol. 32, no. 3, pp. 276–281, 1984.

[6] S. Lopez-Pena and J. R. Mosig, "Analytical evaluation of the quadruple static potential integrals on rectangular domains to solve 3-D electromagnetic problems," *IEEE Transactions on Magnetics*, vol. 45, no. 3, pp. 1320–1323, 2009.

[7] M. G. Duffy, "Quadrature over a pyramid or cube of integrands with a singularity at a vertex," *SIAM Journal on Numerical Analysis*, vol. 19, no. 6, pp. 1260–1262, 1982.